Analogical Extension from Irregular Paradigms in Iranian Armenian: A Case of "Elsewhere Reversal"

Hossep Dolatian and Jordan Kodner
Department of Linguistics and Institute for Advanced Computational Science
Stony Brook University
Stony Brook, NY 11790 USA

February 20, 2026

Abstract

The verbal paradigm of Iranian Armenian has undergone a striking case of analogical extension, separating it from conservative varieties of Eastern Armenian. The past perfective pattern that was historically restricted to an irregular subset of one class of verbs has spread to become the default pattern in the language, while the former default is now restricted to a small class. Since what could be described as the default or elsewhere condition in conservative Eastern Armenian has apparently swapped roles with an irregular pattern, we dub this change an "Elsewhere Reversal." We combine a theoretical analysis with a survey of Armenian dialects and a quantitative model of child language acquisition in order to explain the Elsewhere Reversal. In doing so, we argue that paradoxical cases of dramatic analogical extension such as this can be subsumed under typical mechanisms of analogical leveling.

Keywords: Armenian, Iranian Armenian, morphology, analogical change, child language acquisition, actuation, quantitative modeling, Tolerance Principle

1 Introduction

Our focus in this paper is on the analogical extension of a minority past perfective pattern to the majority class of verbs in the variety of Eastern Armenian spoken in Tehran, Iran (henceforth Iranian Armenian or IA). Both this innovative language variety and the conservative variety of Eastern Armenian (Standard Eastern Armenian (SEA)) have two synthetic past constructions: the subjunctive past and the past perfective, which are cognate across the dialects. They are built on a stem consisting of a root and theme vowel, followed by tense/aspect marking and person/number marking. Verbs are divided into two conjugations, a large *E-Class* and smaller *A-Class*, according to their theme vowel.

In conservative Standard Eastern Armenian, the past perfective is formed with a suffix /-t͡sʰi-/ for all but a handful of irregular verbs which instead take /-∅-ɑ/. The template /-t͡sʰi-/ is the majority form, and so we treat it as the default or elsewhere form. However, the situation appears to be turned on its head in Iranian Armenian. The cognate /-∅-ɒ/ has analogized to the entire E-Class past perfective, restricting /-t͡sʰi-/ to the A-Class. Because the majority form /-t͡sʰi-/ has swapped places with the conditioned /-∅-ɒ/ form in Iranian Armenian, we call this an "Elsewhere Reversal." This is summarized in Table 1, where segmentations are based on our analysis provided in Section 2.[1] In contrast, the subjunctive past is largely identical across the dialects, with a morphophonological difference in the use of theme vowels in hiatus.

Table 1: Overview of synthetic past formation in Standard Eastern Armenian and Iranian Armenian. 3PL forms are provided for concrete illustration.

	Subjunctive past 3PL		Past perfective 3PL	
	SEA	IA	SEA	IA
A-Class 'to read'	kɑɾtʰ-ɑj-i-n	kɒɻtʰ-ɒj-i-n	kɑɾtʰɑ-t͡sʰ-i-n	kɒɻtʰ-ɒ-t͡sʰ-i-n
E-Class 'to sing'	jeɾkʰ-ej-i-n	jeɻkʰ-∅-i-n	jeɾkʰ-e-t͡sʰ-i-n	jeɻkʰ-∅-ɒ-n
Suppletive 'to eat'	ut-ej-i-n	ut-∅-i-n	keɾ-∅-∅-ɑ-n	keɻ∅-∅-ɒ-n
	√-TH-PST-3PL		√-TH-PFV-PST-3PL	

Quantification reveals the Elsewhere Reversal to be a particularly dramatic case of analogical extension. A pattern that only reliably applied to about 30 verbs has leapfrogged its to about 80% of all verbs. Developing an explanation for how the reversal emerged stands to improve not only our understanding of Iranian Armenian, but the nature of analogical extension in general. We approach the problem from three complementary directions. First, in Section 2, we introduce a synchronic theoretical analysis of both conservative Standard Eastern Armenian and innovative Iranian Armenian. The analysis provides an implementation for the change in the grammar and gives it the name "Elsewhere Reversal." This makes sense of the ultimate distribution of the innovative forms in the Iranian Armenian verbal system. This analysis provides enough background to lay out our proposal in Section 3.

Second, we describe cross-dialectal variation in the Armenian synthetic past forms with a focus on Eastern Armenian in 4.3. We find that both the past perfective and the other synthetic form, the subjunctive past, have undergone various changes across dialects, and that changes to the past perfective follow changes to the subjunctive past. We propose that the changes to the subjunctive past are a necessary but not sufficient condition, perhaps a catalyst, for the Elsewhere Reversal in Armenian.

Third, having provided an implementation and laid out related changes across Armenian dialects, Section 5 introduces a mechanism for the analogical change based on a quantitative model of child language acquisition. We draw on over-generalization, a common phenomenon during morphology acquisition as the source of the original innovation that led to the Elsewhere Reversal. This is modeled concretely with the Tolerance Principle (Yang 2005, 2016), a model of productivity learning, which serves as a mechanism for the actuation of the Elsewhere Reversal which makes specific predictions about the conditions under which it could occur. Change to the subjunctive past opened up an alternative generalization over the data that could lead a learner to produce a small extension. Further cohorts of children receiving input from the original innovator(s) could rapidly push the extension forward until it reached the full scope of the Elsewhere Reversal. We discuss how a concrete mechanism for actuation provides explanatory

[1] The SEA segments /ɾ/ and /ɑ/ correspond to IA /ɻ/ and /ɒ/. We will use /A/ to collapse /ɑ/ and /ɒ/ in prose.

power surpassing descriptive tendencies of analogy, such as those related to frequency, paradigm uniformity, or the privileged status of the third person singular.

1.1 Classification and Status of Iranian Armenian

Armenian, attested in the form of Classical Armenian as early as the 5th century AD, is typically classified as a single Indo-European language with multiple varieties, some of which are mutually intelligible (Adjarian 1909; Vaux 1998: ch. 1.1; Dolatian 2024). Contemporary varieties are conventionally divided into two branches, Western and Eastern, each with one standard and a host of non-standard varieties. Iranian Armenian and Standard Eastern Armenian are two mutually intelligible Eastern dialects which share significant similarities in their phonology, morphology, and syntax.

Standard Eastern Armenian (SEA; also called Eastern Armenian) is the primary variety spoken in the modern Republic of Armenia and in diaspora communities in Russia, the Republic of Georgia, the United States, and Iran. The Armenian community in Iran's capital, Tehran, is socially triglossic between Persian, SEA, and Iranian Armenian. Because Persian is the lingua franca in Tehran and national official language, Tehrani Armenians speak Persian when interacting with others who are not Armenian speakers. Within the Armenian community, Standard Eastern Armenian is taught at schools and acts as the formal register of communication. It is the language of education, newspapers, formal speech, so essentially all written materials are in SEA. Iranian Armenian (IA) [iɻɒn-ɑ-hɒjeɻen], also called Persian Armenian [pɒɻsk-ɑ-hɒjeɻen], is the native home language of most Tehrani Armenians. This variety is the one that children acquire at home and that is spoken in informal contexts. Outside of Tehran, it is spoken by the Tehrani Armenian diaspora in the United States, particularly in the Iranian Armenian community of Los Angeles, California. Our data on Iranian is based on fieldwork with speakers from both the Tehran and Los Angeles communities and on a recent Iranian Armenian grammar (Dolatian et al. 2023).

While Armenians have had cultural relations with Persia since antiquity, the modern community in Iran started to develop significantly in the 17th century, due to forced migrations from historic Eastern Armenia, especially Nakhichivan, now part of the Republic of Azerbaijan (Nercissians 2012: 42ff; Barry 2018: 4ff). Some of the largest concentrations of Armenians in Persia first developed in Tabriz, Isfahan, and Salmast, though many speakers eventually migrated to Tehran, during the early-to-mid 20th century. Given this history of migrations, the spoken vernacular of Tehrani Armenian must have leveled into what is now called Iranian Armenian around this time. Although there are many Armenian communities and varieties in modern Iran, the "standardization of the Tehrani dialect has gained much momentum in the last few decades," which is consistent with Iran-internal migrations into Tehran (Nercissians 2001: 67). A large portion of the Tehrani population then emigrated abroad, particularly to California following the 1979 Islamic Revolution. Given the mutual intelligibility between SEA and IA, speakers of either variety often report that IA is just a dialect of Eastern Armenian (Dolatian et al. 2023: 3). This is consistent with the relatively short history of the modern Tehrani variety.

2 A Theoretical Analysis of the "Elsewhere Reversal"

In this section, we provide a formal synchronic analysis of the Standard Eastern Armenian and Iranian Armenian verbal paradigms as groundwork for understanding the Elsewhere Reversal. Our focus is on the form of the synthetic past, but we reference other parts of the paradigm to the extent that they are useful for exposition. For concreteness and clarity, we adopt representational conventions from morpheme-based and item-and-arrangement models like Distributed Morphology (Embick 2015, Halle & Marantz 1993), though our synchronic analysis could reasonably be reformulated in other models (Hockett 1942), whether process-based (Anderson 1992, Aronoff 1976, Stump 2001) or paradigm-based (Blevins 2006).

Consistent with our item-and-arrangement approach, we provide a relatively fine-grained morpheme segmentation throughout this paper. The particulars of our segmentation follow those provided in the grammar Dolatian et al. (2023), the only published grammar on Iranian Armenian. The morphemic structure for verbs was developed by Dolatian and colleagues in earlier work on Standard Western Armenian (Dolatian & Guekguezian 2022a,b, Guekguezian & Dolatian 2024, Karakaş et al. 2021), and later also used to model a wide range of Armenian dialects (Dolatian 2024). This segmentation is more fine-grained than the one found in some older grammars of Armenian (Dum-Tragut 2009), which do not, for example, segment theme vowels from person/number and tense agreement. We note that the grammar provided in Dolatian et al. (2023) is broadly consistent with other recent formal analyses of Armenian morphology, which also segment theme vowels followed by additional tense/aspect/mood and person/number marking (Balabanian 2024, Bezrukov 2022, Khanjian 2013, Vaux 1998). Since the paradigms of Iranian Armenian are described in full in the resources which we cite,[2] we only provide a few relevant paradigms in full in this work.

2.1 Complex and Simplex Verbs

Across Eastern Armenian, verbs can be classified into two morphological categories: simplex and complex. In their citation forms, simplex verbs consist of a root ($\surd$), theme vowel (TH), and infinitival suffix *-l* (INF). Complex verbs contain additional valency-changing derivational morphemes. Simplex verbs fall into two regular conjugations that are distinguished by their theme vowel: the A-Class, which takes the theme vowel *-A-* and the E-Class, which takes *-e-*. The vast majority of verbs (about 80%) belong to the E-Class. We illustrate the division of simplex verbs in Table 2.

Table 2: Overview of simplex verbs in SEA and IA

	Regular simplex		Irregular simplex		
	E-Class	A-Class	Suppletive	Nasal-infixed	
Eastern	jeɾkʰ-e-l	kɑɾtʰ-ɑ-l	ut-e-l	mer-n-e-l	$\surd$-(VX)-TH-INF
Iranian	jeɻkʰ-e-l	kɒɻtʰ-ɒ-l	ut-e-l	mer-n-e-l	$\surd$-(VX)-TH-INF
	'to sing'	'to read'	'to eat'	'to die'	

We describe two types of E-Class verbs as "irregular." The first type consists of suppletive

[2] https://github.com/jhdeov/iranian_armenian/tree/main/VerbConjugation

verbs like ‘to eat’, represented in Tables 2 and 3. These have a distinct present stem used in contexts such as the infinitive, and also an aorist stem used in forms such as the resultative participle and past perfective.[3] The other irregular type, the nasal-infixed verbs, consists of verbs which show a nasal suffix *-n-* (VX) between the root and theme vowel. While this VX appears to be meaningless in the current grammar, it is likely a reflex of the Proto-Indo-European nasal infix (Greppin 1973, Hamp 1975, Kim 2018, Kocharov 2019). The class’s synchronic irregularity comes from the distribution of the nasal suffix, which is deleted in a range of morphological contexts, such as the resultative participle and past perfective. These are thus differentiated from verbs whose roots happen to end in an /-n/ which appears invariantly. The paradigmatic distribution of suffixed and non-suffixed forms of nasal-infixed verbs is similar but not identical to the distribution of present and aorist stems among suppletive verbs(Dolatian & Guekguezian 2022a).

Table 3: Irregular simplex verbs in IA and SEA

	E-Class ‘to mix’		Suppletive ‘to eat’		Nasal-infixed ‘to die’		
	SEA	IA	SEA	IA	SEA	IA	
Infinitive	χɑrn-e-l	χɒrn-e-l	ut-e-l	ut-e-l	mer-n-e-l	mer-n-e-l	√-(VX)-TH-INF
Result. ptcp	χɑrn-ɑt͡s	χɒrn-ɒt͡sʰ	keɾ-ɑt͡s	keɻ-ɒt͡sʰ	mer-ɑt͡s	mer-ɒt͡sʰ	√-RPTCP

Complex verbs extend the root with additional valency-changing affixes, which select their own theme vowels (Table 4): Passives are marked with a *-v-* suffix (PASS) and the *-e-* theme vowel. Causatives are marked with *-t͡sʰn-* (CAUS) along with the *-e-* theme. Sometimes a root’s inherent theme vowel precedes this causative suffix. Inchoatives are marked with the suffix *-n-* (INCH) and the *-A-* theme vowel. INCH is also preceded by *-A-*, which is typically analyzed as a meaningless linking vowel (LV). Many but not all causative and inchoative verbs are synchronically derivable from existing simplex verbs.

Table 4: Synchronically derivable simplex-complex pairs in IA and SEA

	Passivization		Causativization		Inchoativization	
	Base	Passive	Base	Causative	Base	Inchoative
Eastern	jeɾkʰ-e-l	jeɾkʰ-v-e-l	sovoɾ-e-l	sovoɾ-e-t͡sʰn-e-l	uɾɑχ	uɾɑχ-ɑ-n-ɑ-l
Iranian	jeɻkʰ-e-l	jeɻkʰ-v-e-l	sovoɻ-e-l	sovoɻ-e-t͡sʰn-e-l	uɻɒχ	uɻɒχ-ɒ-n-ɒ-l
	‘to sing’	‘to be sung’	‘to learn’	‘to teach’	‘happy’	‘to become happy’
	√-TH-INF	√-PASS-TH-INF	√-TH-INF	√-TH-CAUS-TH-INF	√	√-LV-INCH-TH-INF

2.2 Subjunctive Past

Both Iranian Armenian and Standard Eastern Armenian utilize two synthetic past forms: the subjunctive past (also called the subjunctive past imperfective) and the indicative past perfective (also called the aorist). The subjunctive past has various uses (Dum-Tragut 2009: 239ff; Dolatian et al. 2023: 132). We provide the full paradigm for the two classes for both varieties in

[3] The resultative participle suffix (RPTCP) is *-ɑt͡s* in Eastern with a final voiceless unaspirated affricate. In Iranian, this suffix is prescriptively *-ɒt͡s* with a final voiceless sound as well, but some speakers preferred aspirating this segment.

Table 5. Most agreement (AGR) morphs are identical across both classes and varieties, so going forward, we will only provide the 3PL for illustration.[4]

Table 5: Paradigm of the subjunctive past in Iranian and Eastern

	E-Class ‘to sing’		A-Class ‘to read’	
	SEA	IA	SEA	IA
1SG	jeɾkʰ-ej-i-∅ ‘(if) I were singing’	jeɻkʰ-∅-i-m	kaɾtʰ-aj-i-∅ ‘(if) I were reading’	kɒɻtʰ-ɒj-i-m
2SG	jeɾkʰ-ej-i-ɾ	jeɻkʰ-∅-i-ɻ	kaɾtʰ-aj-i-ɾ	kɒɻtʰ-ɒj-i-ɻ
3SG	jeɾkʰ-e-∅-ɾ	jeɻkʰ-e-∅-ɻ	kaɾtʰ-a-∅-ɾ	kɒɻtʰ-ɒ-∅-ɻ
1PL	jeɾkʰ-ej-i-ŋkʰ	jeɻkʰ-∅-i-ŋkʰ	kaɾtʰ-aj-i-ŋkʰ	kɒɻtʰ-ɒj-i-ŋkʰ
2PL	jeɾkʰ-ej-i-kʰ	jeɻkʰ-∅-i-kʰ	kaɾtʰ-aj-i-kʰ	kɒɻtʰ-ɒj-i-kʰ
3PL	jeɾkʰ-ej-i-n	jeɻkʰ-∅-i-n	kaɾtʰ-aj-i-n	kɒɻtʰ-ɒj-i-n
	√-TH-PST-AGR	√-TH-PST-AGR	√-TH-PST-AGR	√-TH-PST-AGR

The overt past marker /-i/ is consistently used for all Eastern Armenian verbs, including complex verbs and irregular verbs. As shown in Table 6, SEA and IA consistently apply different vowel hiatus repair strategies in all these cases. The /i/ triggers glide epenthesis after /-A-/ in both dialects. But after /e/, we see glide epenthesis in SEA but theme vowel deletion in IA. When the /i/ is absent in the 3SG, the theme vowel surfaces faithfully without any deletion or epenthesis. We argue that they are nevertheless structurally identical, applying a subjunctive past suffix /-i/ after the verb stem and before agreement.

Table 6: Consistent past marker /-i/ for the subjunctive past 3PL throughout Eastern and Iranian Armenian

	SEA	IA	
Causative ‘to teach’	sovoɾ-e-t͡sʰn-e-l sovoɾ-e-t͡sʰn-ej-i̲-n	sovoɻ-e-t͡sʰn-e-l sovoɻ-e-t͡sʰn-∅-i̲-n	√-TH-CAUS-TH-INF √-TH-CAUS-TH-PST-3PL
Passive ‘to be sung’	jeɾkʰ-v-e-l jeɾkʰ-v-ej-i̲-n	jeɻkʰ-v-e-l jeɻkʰ-v-∅-i̲-n	√-PASS-TH-INF √-PASS-TH-PST-3PL
Inchoative ‘to become happy’	uɾaχ-a-n-a-l uɾaχ-a-n-aj-i̲-n	uɻɒχ-ɒ-n-ɒ-l uɻɒχ-ɒ-n-ɒj-i̲-n	√-LV-INCH-TH-INF √-TH-PST-3PL
Infixed ‘to die’	mer-n-e-l mer-n-ej-i̲-n	mer-n-e-l mer-n-∅-i̲-n	√-VX-TH-INF √-VX-TH-PST-3PL
Suppletive ‘to eat’	ut-e-l ut-ej-i̲-n	ut-e-l ut-∅-i̲-n	√-TH-INF √-TH-PST-3PL

In sum, the past marker is /-i/ is uniformly applied for all conjugation classes in the subjunctive past. This is accounted for by the rule in (1), in which we treat /-i/ as a spellout of

[4] One exception is the past 1SG marker, which is covert in SEA and *-m* in IA.[5] The past marker is covert in the 3SG, overt *-i* elsewhere. Agreement marking is essentially the same between the subjunctive past and the past perfective except for the 3SG. In the past perfective 3SG, the ending is TH-/t͡sʰ-∅-∅/ or /∅-A-v/ depending on the aspect/tense allomorphy described below.

(T)ense [6]

(1) *Rules for past tense in the subjunctive past*
T[+PST] ↔ *-i* / Elsewhere

2.3 Simplex Past Perfective

Standard Eastern Armenian and Iranian Armenian differ more significantly in how they express their past perfective. SEA uses the thematic perfective-past sequence of /-t͡sʰ-i/ as its apparent default template, while an athematic template /-∅-ɑ/ is mainly relegated to suppletives and nasal-infix verbs. In contrast, IA uses the /-∅-ɒ/ as its default, while /-t͡sʰ-i/ is restricted to the A-Class. Thus, the past marker in the past perfective is either /-i-/ or /-A-/, and the perfective marker is either zero -∅ or an affricate /-t͡sʰ-/. We illustrate in Table 7 with an A-Class verb, a suppletive verb, and an infixed verb, since these three categories do not change in IA. We underline the perfective and past morphs and use ∅ here to annotate deleted or absent morphs when compared with the infinitive, such as the deleted theme vowels and nasal suffix.

Table 7: Templates for past perfective marking in A-Class and irregular infixed verbs

	SEA	IA	
A-Class 'to read'	kɑɾtʰ-ɑ-l	kɒɻtʰ-ɒ-l	√-TH-INF
'they read (PST)	kɑɾtʰ-ɑ-t͡sʰ-i-n	kɒɻtʰ-ɒ-t͡sʰ-i-n	√-TH-PFV-PST-3PL
Suppletive 'to eat'	ut-e-l	ut-e-l	√-TH-INF
'they ate'	keɾ-∅-∅-ɑ-n	keɻ-∅-∅-ɒ-n	√-TH-PFV-PST-3PL
Irregular infixed 'to die'	mer-n-e-l	mer-n-e-l	√-VX-TH-INF
'they died'	mer-∅-∅-∅-ɑ-n	mer-∅-∅-∅-ɒ-n	√-VX-TH-PFV-PST-3PL

As a rule (2), the template /-∅-A/ triggers the deletion of any preceding theme vowels. The nasal suffix is likewise deleted before this suffix. The resulting adjacency of the Asp[PFV]-T[PST] complex to the root further allows it to select the aorist root for suppletive verbs, which is blocked in the subjunctive past by the intervening theme vowel.

(2) *Readjustment rule*: *Delete theme vowels and nasal suffix before the* Asp[PFV]-T[PST] /-∅-A/ *complex*
/e_{th}/ → ∅ / _ A
/n_{vx}/ → ∅ / _ A
(where /e/ is a theme vowel TH, /n/ is the nasal suffix VX,
and /A/ is a past marker PST: SEA /ɑ/, IA /ɒ/)

Both varieties make some use of the /-t͡sʰ-i/ and /-∅-A/ patterns for perfective marking, but their distributions differ in the E-Class (Table 8). In Standard Eastern Armenian, /-t͡sʰ-i/ predominates outside the irregulars, while in Iranian Armenian, /-∅-A/ appears consistently instead.

[6] In the 3SG subjunctive past, the past marker is zero: [jeɻkʰ-e-∅-ɻ] 'if he were singing'. This morph is tangential to our eventual discussion of the past perfective, so we set it aside. See Karakaş et al. (2021) for the appropriate rules based on the Western Armenian cognates.

Table 8: Different predominant patterns for past perfective marking in the E-Class

	SEA	IA	
E-Class 'to sing'	jeɾkʰ-e-l	jeɻkʰ-e-l	$\sqrt{}$-TH-INF
'they sang'	jeɾkʰ-e-<u>t͡sʰ-i</u>-n	jeɻkʰ-∅-<u>∅-ɒ</u>-n	$\sqrt{}$-TH-<u>PFV-PST</u>-3PL

The generalization is that in Standard Eastern, the default template (i.e., the elsewhere pattern) for the indicative past perfective is /-t͡sʰ-i/, while it is /-∅-A/ in Iranian. In the subjunctive past, the past is uniformly marked as /-i/ in both dialects. We formalize this state of affairs below. For illustrative purposes, we use realization rules for the entire template -ASP[PFV]-T[PST] (3). In SEA, this template is realized as /-∅-A/ (where /A/ is /ɑ/) for a handful of irregular verbs like 'to eat', while it is /-t͡sʰ-i/ elsewhere for the E-Class and A-Class. In contrast in IA, the template /-t͡sʰ-i/ is limited to the A-Class, and /-∅-A/ (where /A/ is /ɒ/) is the default pattern which applies when no other conditions are met. Since the two past perfective patterns have effectively swapped places between conditioned and elsewhere, we call this the Elsewhere Reversal.[7]

(3) The Elsewhere Reversal: Exponence rules for the template -ASP[PFV]-T[PST] in the past perfective for the E-Class, A-Class, suppletive 'to eat', and infixed verbs like 'to die'

Standard Eastern Armenian

PFV-PST	→	-∅-ɑ	/ {$\sqrt{\text{EAT}}$, $\sqrt{\text{DIE}}$, ...} _
	→	-t͡sʰ-i	/ Elsewhere

Iranian Armenian

PFV-PST	→	-t͡sʰ-i	/ TH[=ɒ] _
	→	-∅-ɒ	/ Elsewhere

We follow previous work on Armenian in treating /-t͡sʰ/ as a perfective marker (Donabedian 2016), and in treating the vowels /-i, -A/ as past tense allomorphs (Dolatian & Guekguezian 2022b, Karakaş et al. 2021). The past perfective Elsewhere Reversal in conjunction with vowel deletion in the subjunctive past yields a surface generalization in the Iranian Armenian E-Class: the two synthetic conjugations are only distinguished by the quality of their vowel (e.g., subjunctive past *jeɻkʰ-i-n*, past perfective *jeɻkʰ-ɒ-n*). This proves critical to our analysis of the diachronic process of the Elsewhere Reversal in Sections 4.3 and 5.

2.4 Complex Past Perfective

The Elsewhere Reversal affects both simplex verbs and complex verbs in Iranian Armenian. Since passives take the /-e-/ theme vowel in Eastern Armenian, the Elsewhere Reversal affects them as it does the simplex E-Class. The past perfective template is /-t͡sʰ-i/ in Standard Eastern Armenian, while /-∅-A/ (/-∅-ɒ/) for Iranian Armenian (Table 9).

[7]These rules ignore complications from the 3SG, since they do not affect the core generalizations relevant to this article. For an analysis of the covert past and 3SG markers, see Karakaş et al. (2021).

Table 9: Different predominant patterns for past perfective marking in passives

Passive	SEA	IA	
‘to be sung’	jeɾkʰ-v-e-l	jeɻkʰ-v-e-l	√-PASS-TH-INF
‘they were sung’	jeɾkʰ-v-e-t͡sʰ-i-n	jeɻkʰ-v-∅-∅-ɒ-n	√-PASS-TH-PFV-PST-3PL

Matters are slightly more complicated for causatives (Table 10). The causative suffix is /-t͡sʰn-/ in its elsewhere form, but /-t͡sʰɾ/ (SEA) and /-t͡sʰɻ/ (IA) in the past perfective. In Iranian Armenian, the past perfective ending is consistently /-∅-A/ as it is for passives, but in Standard Eastern Armenian, there is a register distinction. In colloquial speech, causatives pattern like simplex E-Class verbs. However, in formal contexts, the expected /-e-t͡sʰ-/ is omitted while retaining /-i/ for tense.

Table 10: Different predominant patterns for past perfective marking in causatives

Causative		SEA	IA	
‘to teach’		sovoɾ-e-t͡sʰn-e-l	sovoɻ-e-t͡sʰn-e-l	√-TH-CAUS-TH-INF
‘they taught’	Std.	sovoɾ-e-t͡sʰɾ-∅-∅-i-n	sovoɻ-e-t͡sʰɻ-∅-∅-ɒ-n	√-TH-CAUS-TH-PFV-PST-3PL
	Coll.	sovoɾ-e-t͡sʰɾ-e-t͡sʰ-i-n		

Inchoatives, which take the theme vowel /-A-/, remain the odd ones out in the past perfective. In both dialects, they share the overt perfective /-t͡sʰ-/ typical of A-Class verbs, but they drop their inchoative affix and mark tense with /-A/ (Table 11).

Table 11: Past perfective template for inchoatives in Eastern and Iranian

Inchoative	SEA	IA	
‘to become happy’	uɾɑχ-ɑ-n-ɑ-l	uɻɒχ-ɒ-n-ɒ-l	√-LV-INCH-TH-INF
‘they became happy’	uɾɑχ-ɑ-∅-∅-t͡sʰ-ɑ-n	uɻɒχ-ɒ-∅-∅-t͡sʰ-ɒ-n	√-LV-INCH-TH-PFV-PST-3PL

The general morphological and semantic homogeneity of this class allows us to condition this behavior on the INCH morpheme (Dolatian & Guekguezian 2022a). The inchoative morpheme is covert, and exceptionally selects /-t͡sʰ-A/.

2.5 Irregular Verbs and Lexical Exceptions to the Synthetic Past

A handful of verbs exist as exceptions to the generalizations summarized in Tables 7 and 11. Their behavior in both Standard Eastern Armenian and Iranian Armenian support our analysis of the analogical extension in Iranian Armenian as the Elsewhere Reversal. As stated in §2.1, irregular verbs tend to take the past perfective template /-∅-A/. We list suppletive verbs following this pattern in Table 12.

Table 12: Suppletive verbs that take /-∅-A/ for the past perfective in Eastern and Iranian

	SEA	IA	
‘to eat’	ut-e-l	ut-e-l	√-TH-INF
‘they ate’	keɾ-∅-∅-ɑ-n	keɻ-∅-∅-ɒ-n	√-TH-PFV-PST-3PL
‘to come’	ɡ-ɑ-l	ɡ-ɒ-l	√-TH-INF
‘they came’	jek-∅-∅-ɑ-n	ek-∅-∅-ɒ-n	√-TH-PFV-PST-3PL
‘to take to’	tɑn-e-l	tɒn-e-l	√-TH-INF
‘they took’	tɑɾ-∅-∅-ɑ-n	tɒɻ-∅-∅-ɒ-n	√-TH-PFV-PST-3PL

The verb ’to be’ occupies different irregular categories between SEA and IA, and it keeps the past perfective template /-∅-A/. It is a suppletive verb [lin-e-l] in SEA, while it is expressed by an irregular infixed verb in IA [el-n-e-l] (Table 13).

Table 13: Irregular verb ‘to be’ in the past perfective for Eastern and Iranian

	SEA	IA	
‘to be’	lin-e-l	el-n-el	√-(VX)-TH-INF
‘they were’	jeʁ-∅-∅-ɑ-n	el-∅-∅-∅-ɒ-n	√-(VX)-TH-PFV-PST-3PL

A handful of Standard Eastern Armenian infixed and suppletive verbs express the Regular E-Class /-t͡sʰ-i/ instead of the expected /-∅-A/ (Table 14). This includes nasal-infixed ‘to let’ and three suppletive verbs ‘to give’, ‘to put’, and ‘to do’. Each of these “irregular irregulars” has adopted /-∅-A/ in IA, bringing them in line with the rest of the E-Class.

Table 14: Exceptional templates for past perfective marking in irregular infixed and suppletive verbs

	SEA	IA	
‘to let’	tʰoʁ-n-e-l	tʰoʁ-n-e-l	√-VX-TH-INF
‘they let (PST)’	tʰoʁ-∅-e-t͡sʰ-i-n	tʰoʁ-∅-∅-ɒ-n	√-VX-TH-PFV-PST-3PL
‘to give’	t-ɑ-l	t-ɒ-l	√-TH-INF
‘they gave’	təv-e-t͡sʰ-i-n	təv-∅-∅-ɒ-n	√-TH-PFV-PST-3PL
‘to put’	dən-e-l	dən-e-l	√-TH-INF
‘they put (PST)’	dəɾ-e-t͡sʰ-i-n	dəɻ-∅-∅-ɒ-n	√-TH-PFV-PST-3PL
‘to do’	ɑn-e-l	ɒn-e-l	√-TH-INF
‘they did’	ɑɾ-e-t͡sʰ-i-n	ɒɻ-∅-∅-ɒ-n	√-TH-PFV-PST-3PL

A few verbs only express irregularity in the imperative 2SG. These verbs express it with a bare root or some irregular suffix rather than the expected *-iɾ* suffix (Table 15). Such verbs take the /-t͡sʰ-i/[8] template for the past perfective in Standard Eastern Armenian and the /-∅-A/ template in IA.

[8]In SEA, the verb ‘to say’ irregularly takes a theme vowel /ɑ/ in the past perfective.

Table 15: Irregular verbs that are irregular in imperative 2SG but not in their past perfective template

	SEA	IA	
‘to say’	ɑs-e-l	ɒs-e-l	√-TH-INF
‘they said’	ɑs-ɑ-t͡sʰ-i-n	ɒs-∅-∅-ɒ-n	√-TH-PFV-PST-3PL
‘say! (2SG)”	ɑs-ɑ	ɒs-ɒ	√-TH
‘to bring’	beɽ-e-l	beɻ-e-l	√-TH-INF
‘they brought’	beɾ-e-t͡sʰ-i-n	beɻ-∅-∅-ɒ-n	√-TH-PFV-PST-3PL
‘bring! (2SG)’	beɾ	beɻ	√

Our analysis of the Elsewhere Reversal neatly accounts for the behavior of these verbs in both dialects. In Standard Eastern Armenian, the exceptional irregular verbs can be analyzed as missing from the list of conditioning roots for /-∅-A/ in (3). In Iranian Armenian, /-∅-A/ applies to these by default as a consequence of the Elsewhere Reversal. In diachronic approaches to DM and similar frameworks, this would be considered a case of analogical leveling (Hock 2003) implemented as rule impoverishment (Calabrese & Grestenberger 2024, Kodner 2023b).

Colloquial Eastern Armenian (CEA) and Iranian Armenian share one A-Class suppletive verb, ‘to go’ (Table 16). It is inflected as A-Class verb with /-t͡sʰ-i/ in both dialects.

Table 16: Verb ‘to go’ and its past perfective in both Eastern and Iranian

	SEA	CEA	IA	
‘to go’	gən-ɑ-l	jeɾtʰ-ɑ-l	e(ɻ)tʰ-ɒ-l	√-TH-INF
‘they went’	gən-ɑ-t͡sʰ-i-n	gən-ɑ-t͡sʰ-i-n	gən-ɒ-t͡sʰ-i-n	√-TH-PFV-PST-3PL

Finally, there is one suppletive inchoative verb ‘to turn into’ (Table 17). This behaves identically to other inchoatives in both dialects, because the exponent /t͡sʰ-A/ is conditioned on INCH. Note the suppletive change in the root-final rhotic. These are not allophones.

Table 17: Irregular inchoative with a changing liquid and their past perfective template

	SEA	IA	
‘to turn into’	dɑr-n-ɑ-l	dɒr-n-ɒ-l	√-INCH-TH-INF
‘they turned into’	dɑɾ-∅-∅-t͡sʰ-ɑ-n	dɒɻ-∅-∅-t͡sʰ-ɒ-n	√-INCH-TH-PFV-PST-3PL

Finally, this contrasts with a second irregular inchoative ‘to wash’ (Table 18). This verb has a surface pattern identical to simplex A-Class verbs in both dialects. The verb is additionally exceptional in that it is morphologically inchoative (based on its other inflected forms outside of the past perfective) yet is semantically transitive.

Table 18: Past perfective template for irregular inchoatives in Eastern and Iranian

Irregular inchoative	SEA	IA	
‘to wash’	ləv-ɑ-n-ɑ-l	ləv-ɒ-n-ɒ-l	√-LV-INCH-TH-INF
‘they washed’	ləv-ɑ-∅-∅-t͡sʰ-i-n	ləv-ɒ-∅-∅-t͡sʰ-i-n	√-LV-INCH-TH-PFV-PST-3PL

2.6 Quantitative Summary

Iranian Armenian has drastically altered the distribution of inflectional patterns in the past perfective system. Compared to the conservative Standard Eastern Armenian dialect, it has significantly reduced the distribution of the template /-t͡sʰ-i/, having demoted it from the elsewhere form to one conditioned on inflectional class. In contrast, it has promoted /-∅-A/ from the scope of a handful of irregular verbs into the elsewhere form, greatly widening its distribution. A quantitative analysis of class sizes emphasizes just how drastic this extension really was: over 80% of Standard Eastern Armenian verbs shifted from taking /-t͡sʰ-i/ to /-∅-A/ in Iranian Armenian.

Table 19 provides estimates of verb class sizes and summarizes the patterns and scope of the Elsewhere Reversal extension discussed in this section. The numbers are taken from the lemma count of these verbs in the lexicon of the Eastern Armenian National Corpus (EANC; Khurshudian et al. 2009)[9]

[9]The lexicon was accessed (Sept 2020) from EANC's source-code from the verb lexeme list: `https://bitbucket.org/timarkh/uniparser-grammar-eastern-armenian/src/master/hye-lexemes-V.txt`. We used the following EANC paradigm indices to find the relevant classes: irregular infixed (V12_main or V13), causative (V14_main), A-Class (V21), and inchoative (V22_main). Passives are in the V11_main paradigm, end in վել, and have the 'med' tag. The E-Class is all other members of the V11_main paradigm.

Table 19: Distribution of past perfective templates across regular and irregular verbs

Category		Past perfective template SEA	IA	Class size in SEA?	/-∅-A/ in SEA	/-∅-A/ in IA
Regular						
E-Class	(Table 8)	/-t͡sʰ-i/	/-∅-ɒ/	6672 (46.87%)		✓
Passive	(Table 9)	/-t͡sʰ-i/	/-∅-ɒ/	4089 (28.73%)		✓
Causative	(Table 10)	/-∅-i/ (Std.) /-t͡sʰ-i/ (Coll.)	/-∅-ɒ/	1457 (10.24%)		✓
Inchoative	(Table 11)	/-t͡sʰ-ɑ/	/-t͡sʰ-ɒ/	1191 (8.37%)		
A-Class	(Table 7)	/-t͡sʰ-i/	/-t͡sʰ-i/	778 (5.5%)		
Irregular or suppletive						
Infixed	(Table 7)	/-∅-ɑ/	/-∅-ɒ/	28 (0.20%)	✓	✓
‘to let’	(Table 14)	/-t͡sʰ-i/	/-∅-ɒ/	1		✓
‘to come’	(Table 12)	/-∅-ɑ/	/-∅-ɒ/	1	✓	✓
‘to eat’	(Table 1)	/-∅-ɑ/	/-∅-ɒ/	1	✓	✓
‘to take to’	(Table 12)	/-∅-ɑ/	/-∅-ɒ/	1	✓	✓
‘to give’	(Table 14)	/-t͡sʰ-i/	/-∅-ɒ/	1		✓
‘to put’	(Table 14)	/-t͡sʰ-i/	/-∅-ɒ/	1		✓
‘to do’	(Table 14)	/-t͡sʰ-i/	/-∅-ɒ/	1		✓
‘to go’	(Table 16)	/-t͡sʰ-i/	/-t͡sʰ-i/	1		
‘to be’	(Table 13)	/-∅-ɑ/	/-∅-ɒ/	1	✓	✓
‘to say’	(Table 15)	/-t͡sʰ-i/	/-∅-ɒ/	1		✓
‘to bring’	(Table 15)	/-t͡sʰ-i/	/-∅-ɒ/	1		✓
‘to turn into’	(Table 17)	/-∅-ɑ/	/-∅-ɒ/	1	✓	✓
‘to wash’	(Table 18)	/-t͡sʰ-i/	/-t͡sʰ-i/	1		
Total				14234		

The EANC lexicon contains 14,329 verb lemmas. Since some irregular and suppletive verbs listed in the EANC are absent from Iranian Armenian, we supplement this with a list of irregular verbs reported for the dialect in (Dolatian et al. 2023). We likewise set aside Eastern verbs which belong to archaic conjugation classes. This leaves 14,234 verbs, which are reported in the table.

3 Proposal for the “Elsewhere Reversal”

As argued, the Elsewhere Reversal is robust in the current synchronic snapshot of Iranian Armenian. But how did this system arise? We argue that the Elsewhere Reversal was partially triggered by other language changes. Specifically, we argue that Elsewhere Reversal depended on a morphophonological change in the subjunctive past. Evidence will come from data across multiple dialects in the next section.

Synthetic past forms in Iranian Armenian show two innovations relative to the conservative Eastern Armenian system represented by Standard Eastern Armenian. First, while the subjunctive past employs the past suffix /-i-/ throughout all conjugational classes in both dialects, Iranian Armenian innovatively deletes the theme vowel /-e-/ before the past suffix /-i-/. The distribution of surface athematic subjunctive pasts lines up with the extent of the analogical extension of /-∅-A/, leading to two parallel changes (bold in Table 20).

Table 20: Absence of theme vowel /e/ before the past suffix in Iranian Armenian with innovative forms bolded

	Subjunctive Past 3PL		Ind. Past Perfective 3PL	
	SEA	IA	SEA	IA
A-Class ‘to read’	kɑɾtʰ-ɑj-i-n	kɒɻtʰ-ɒj-i-n	kɑɾtʰ-ɑ-t͡sʰ-i-n	kɒɻtʰ-ɒj-t͡sʰ-i-n
E-Class ‘to sing’	jeɾkʰ-ej-i-n	**jeɻkʰ-∅-i-n**	jeɾkʰ-e-t͡sʰ-i-n	**jeɻkʰ-∅-∅-ɒ-n**
Suppletive ‘to eat’	ut-ej-i-n	**ut-∅-i-n**	keɾ-∅-∅-ɑ-n	keɻ-∅-∅-ɒ-n
	√-TH-PST-3PL		√-TH-PFV-PST-3PL	

We propose that this morphophonological change to the subjunctive past created the opportunity for the analogical extension in the past perfective. Once the morphophonological change occurred, new generalizations became available in the grammar.

Once the theme vowel was lost in the E-Class subjunctive past, a new surface-true generalization held for the synthetic past of the (often high-frequency) E-Class irregulars, namely that the past is marked athematically and that perfectivity may be indicated by the quality of the vowel (/-i-/ or /-A-/). A speaker’s (or child learner’s) hypothesis that /-i-/ simply indicated past non-perfective would be supported by the complementary distribution of surface /-ɒ/ (PST.PFV) with overt agglutinative /-t͡sʰ-i/ (PFV-PST). This generalization, “*mark past with vowel quality*,” would then be available to extend to other verbs. We apply a quantitative model of child language acquisition to capture this extension and its ultimate distributional limits in Section 5. Crucially, this analogical extension is just a case of leveling from a productive default pattern (a learner’s innovative grammar) to non-productive patterns.

This proposal comes with an additional surface-true generalization regarding the prosodic form of the past. We adopt this throughout the remainder of the paper for purposes of exposition: once the /-e-/ theme vowel was lost in the subjunctive past, both past forms for irregular E-Class verbs with /-∅-∅-A/ past perfectives shared the prosodic pattern *root-VC*, while other E-Class verbs shared *root-VC* in the subjunctive past and *root-VCVC* in the past perfective, and A-Class verbs retained *root-VCVC* in both.[10] This could be described as a paradigm uniformity effect (Kenstowicz 1996).

4 Dialectal variation in the Past Perfective and Subjunctive Past

In this section, we discuss intra- and inter-dialectal variation among past perfective forms within Colloquial Eastern Armenian and its differents from Standard Eastern Armenian. Grammars of Standard Eastern Armenian provide a simplified story, as /-∅-A/ can optionally appear among regular *E*-class verbs in Colloquial Eastern Armenian. If they occurred optionally in Colloquial Eastern Armenian before Iranian Armenian diverged from it, then they may have had a facilitory effect on the Elsewhere Reversal.

That said, our account for the Elsewhere Reversal primarily depends on the emergent prosodic parallelism that emerged with changes to the subjunctive past. To test this proposal, we survey the subjunctive past and past perfective patterns for fifty-three 20th century Armenian dialects. We establish that many dialects occupy positions intermediate to Standard Eastern

[10]This template is violated only in the 3SG for SEA where the past morph and agreement morph are covert, and in the 1SG for SEA where agreement is covert.

Armenian and Iranian Armenian. Among these, it is consistently the case that dialects with at least a partial Elsewhere Reversal also show changes to their subjunctive past. This is consistent with a necessary but not sufficient causal relationship between the two.

4.1 Diachronic origins of the past perfective templates from Classical Armenian

The ancestors of the modern past perfective (aorist) templates /-t͡sʰ-i/ and /-∅-A/ are attested in Classical Armenian. But the template show a semantic distinction indicating voice (Godel 1975, Olsen 2017, Thomson 1989). Briefly, the /-A-/ tends to correlate with passive voice in CA. And separately, regular or 'weak' verbs took the perfective suffix /-t͡sʰ/, while 'strong' or irregular verbs took the zero suffix /-∅/. (Godel 1975: 38,43ff; Thomson 1989: 48,90; Olsen 2017: 1093).

Table 21: The aorist or past perfective in Classical Armenian (Thomson 1989: 48)

	Regular & active 'they liked'	Irregular & active 'they brought'	Regular & passive 'they were liked	Irregular & passive 'they were brought'
IPA	siɾ-e-t͡sʰ-i-n	beɾ-∅-∅-i-n	siɾ-e-t͡sʰ-ɑ-n	beɾ-∅-∅-ɑ-n
	√-TH-PFV-PST-3PL			
	սիրեցին	բերին	սիրեցան	բերան

As the language developed into its modern Armenian, the templates expanded or retracted across different lexemes and conjugation classes (cf. case studies in Martirosyan 2018). In modern varieties, one often finds that the use of the past suffix /-A/ (and past perfective perfective template /-∅-A/), is correlated with intransitivity.

4.2 Variable Past Perfectives Among High-Frequency Verbs in modern Armenian

Across the board, modern Armenian varieties function more similarly to each other than to Standard Eastern Armenian and Iranian Armenian than to Classical Armenian in that they have lost the robust semantic distinction between the two past perfective templates. This motivates our general focus on modern dialects to the exclusion of CA in the remainder of this paper.

The first piece of evidence supporting our hypothesis regarding the link between the changes to the synthetic past comes from the distribution of optional /-∅-A/ in modern dialects, including Standard Eastern Armenian. It is common to find /-∅-A/ as an option for some verbs in their 3SG forms. But notably, these remain a finite set of optional forms with a distribution among high-frequency verbs. In other words, these are consistent with memorized forms that have not entered into the dialects' productive morphology.

This optionality is correlated with a register distinction. Within Standard Eastern Armenian, there are some high-frequency verbs that use the /-t͡sʰ-i/ template in standard speech but exhibit the /-∅-A/ template in colloquial speech, especially in the 3SG (Table 22). In the 3SG, the past marker /-ɑ/ is followed by an overt 3SG marker /-v/. The data below are taken from Dum-Tragut (2009: 230) (citing Ղարագյուլյան 1981: 98). The IPA transcriptions are our own, and the segmentation and inclusion of zero morphs corresponds with our analyses in the previous section.

Table 22: High-frequency verbs that colloquially take /-∅-ɑ/ in Eastern Armenian

Infinitive		Past perfective 3SG Standard Eastern	Colloquial Eastern	
	$\sqrt{}$-TH-INF	$\sqrt{}$-TH-PFV-PST-3SG	$\sqrt{}$-TH-PFV-PST-3SG	
‘to sit’	nəst-e-l	nəst-e-t͡s-∅-∅	nəst-∅-∅-ɑ-v	‘he sat’
‘to bring’	beɾ-e-l	beɾ-e-t͡s-∅-∅	beɾ-∅-∅-ɑ-v	‘he brought’
‘to give’	t-ɑ-l	təv-e-t͡s-∅-∅	təv-∅-∅-ɑ-v	‘he gave’
‘to say’	ɑs-e-l	ɑs-ɑ-t͡s-∅-∅	ɑs-∅-∅-ɑ-v	‘he said’
‘to start’	skəs-e-l	skəs-e-t͡s-∅-∅	skəs-∅-∅-ɑ-v	‘he started’

Avetyan (Ավետյան 2020) provides more examples of high-frequency verbs that show the above variation (Table 23). He also provides data showing that this change is also affecting passive verbs in Colloquial Eastern Armenian. Personal communication with Avetyan suggests that the template has spread quite quickly to all passives and intransitives in colloquial Eastern. Although most attested examples are for the 3SG, Avetyan suggests that the other persons can also take the /-∅-A/ template. The original data is from his paper and personal communication, but IPA and segmentations are our own.[11] We discuss the possible implications of this in Section 4.7.

Table 23: Passives verbs that colloquially take /-∅-ɑ/ in Eastern Armenian

Infinitive		Past perfective 3SG Standard Eastern	Colloquial Eastern	
	$\sqrt{}$-PASS-TH-INF	$\sqrt{}$-PASS-TH-PFV-PST-3SG		
‘to be accepted’	əntʰun-v-e-l	əntʰun-v-e-t͡s-∅-∅	əntʰun-v-∅-∅-ɑ-v	‘they were accepted’
‘to be killed’	spɑn-v-e-l	spɑn-v-e-t͡s-∅-∅	spɑn-v-∅-∅-ɑ-v	‘they were killed’

More distantly related Western Armenian (Table 24) exhibits some similar morphological optionality, but it is more difficult to line these up with the changes in Eastern varieties (Boyacioglu 2010, Boyacioglu & Dolatian 2020). Some high-frequency irregular verbs like ‘to bring’ switch between inflection templates based on specific person-number features (i.e., heteroclisis, Stump 2006). In the case of the past perfective, the Western irregular verb ‘to bring’ takes the /-∅-A/ template for the 3SG, and /-∅-i/ for other persons/numbers. Some verbs like ‘to sit’ take a theme vowel /-i-/ and use the past perfective template /-t͡sʰ-A/ in standard Western, but /-∅-A/ in colloquial speech for all persons/numbers. Furthermore, for colloquial Western Armenian in Lebanon, some passives can take /-∅-A/, especially for the 3SG. Data is from our native judgments.

[11] Because of the lack of written records on colloquial usages, it is unknown when Colloquial Eastern Armenian started using these /-∅-A/ forms. Avetyan inform us that early 20th century Armenian grammars did sometimes mention 3SG forms with the /-∅-A/ template for high-frequency words like ‘to sit’. But it is unknown how pervasive these colloquialisms were.

Table 24: High-frequency verbs that variably take /-∅-ɑ/ in Western Armenian

	Infinitive	Std. 3SG	Coll. 3SG	Std. 3PL	Coll. 3PL
	√-(PASS)-TH-INF	√-(PASS)-TH-PFV-PST-3SG		√-(PASS)-TH-PFV-PST-3PL	
'to bring'	pʰeɾ-e-l	pʰeɾ-∅-∅-ɑ-v	pʰeɾ-∅-∅-ɑ-v	pʰeɾ-∅-∅-i-n	pʰeɾ-∅-∅-i-n
'to sit'	nəst-i-l	nəst-e-t͡s-ɑ-v	nəst-∅-∅-ɑ-v	nəst-e-t͡s-ɑ-n	nəst-∅-∅-ɑ-n
'to be destroyed'	ɑvɾə-v-i-l	ɑvɾə-v-e-t͡s-ɑ-v	ɑvɾə-v-∅-∅-ɑ-v	ɑvɾə-v-e-t͡s-ɑ-n	ɑvɾə-v-∅-∅-ɑ-n

Outside of these colloquial registers, some dialects primarily use the template /-t͡sʰ-i/ for the E-Class and other regular verbs, while intransitives and/or passives use the template /-∅-A/ in a way somewhat reminiscent of Classical Armenian. Tigranekert Armenian is an example of such a dialect (Հանեյան 1978: 111).

Within Iran, Adjarian reports that various Armenian dialects other than Tehrani Armenian have extended the use of the past marker /-A/ to some degree (Աճառյան 1959: 473-475; 1961: 201).[12] He specifically cites the dialects of Khoy, Salmast, Tabriz, and Maragha as examples. It is unclear to what extent this is a shared innovation of the Iranian branch of Armenian that has developed to the most extreme degree in Tehrani, or whether the extension is due to contact between Tehrani and outlying Iranian varieties. Both scenarios would be consistent with our model of analogical extension.

In summary, several Armenian varieties and registers optionally extend the /-∅-A/ template to some classes or sets of verbs. In the case of colloquial Eastern and Western Armenian, this spread has primarily been limited to a small set of irregular, high-frequency, and sometimes passive/intransitive verbs. But in Iranian Armenian, this extension has been widespread, affecting verbs of any valency if they share the /-e-/ theme vowel. This is consistent with our proposal for the Elsewhere Reversal (Section 3), since the mere availability of /-∅-A/ among individual lexical items does not automatically entail its dramatic extension in the grammar.

4.3 Dialect Survey of Partial "Elsewhere Reversals"

We propose that the emergent prosodic parallelism between the innovative single vowel /-i-/ subjunctive forms and the /-∅-A/ past perfective facilitated the Elsewhere Reversal. These are the two basic synthetic past structures in Armenian. To test this proposal, we survey dialectal data on the conjugation of the regular E-Class. One main source is Adjarian 1911 (Աճառեան 1911) which provides sketches on 7 Eastern and 21 Western non-standard varieties.[13] Our primary finding in this section is that almost all dialects with /-∅-A/ or similar non-consonantal past perfective patterns either exhibit subjunctive past forms with a single post-root vowel or have lost the synthetic subjunctive past altogether. However, the converse does not hold (with two possible exceptions §4.5.5) which suggests that the change to the subjunctive past is an important but not sufficient condition for giving rise to the "Elsewhere Reversal" and the Iranian Armenian system.

[12]Some dialectological sources conflate the spread of /-A/ with the use of the zero perfective morph -∅ (Ջահուկյան 1972: p. 103, feature 100.6).

[13]The original Adjarian monograph is in Armenian, but a translation is available (Dolatian 2024). Note that the majority of these early 20th century dialects are currently dead or moribund, as a consequence both of recent dialect leveling and of the Armenian Genocide.

We take the 31 dialects in Adjarian 1911 as well as 22 additional dialects from a range of other sources as a representative sample of 53 Armenian dialects. In the following section, we use asterisks to mark dialects for which we supplemented Adjarian 1911 with other grammars. The other grammars provide information on various subdialects as well, and sometimes a dialect that is not included in Adjarian's original sample. Adjarian 1911 provides uneven detail on these 31 varieties ranging from full paradigms for both synthetic tenses for some to only partial descriptions for others. Throughout the survey, we use boldface to mark the segments in the verb that come after the root in order to highlight the similar or dissimilar prosodic structures. We underline the past and perfective morphs in the verb.

4.4 Dialects that are faithful to the original SEA-like templates: Subjunctive past *root-V(j)VC*, and perfective *root-VCVC*

Of Adjarian's 31 varieties, 14 followed the SEA-like pattern where the past perfective uses the morphological template /-t͡s-i/ and the subjunctive past uses two vowels on the surface /-e-i/, thus creating matching prosodic templates between the two past forms. Note that in following standard Armenian orthographic conventions, dialectological work tends to not transcribe epenthetic glides between the vowels /e/ and /i/. But we can assume that these dialects had an epenthetic glide like contemporary dialects, thus showing the prosodic template *root-VCVC*, or minimally, an open syllable followed by a closed syllable. Crimean Armenian is such a dialect (4).

(4) Crimean Armenian

*siɾ-**e-i-n**,* *siɾ-**e-tsʰ-i-n***
like-**TH-PST-3PL**, like-**TH-PFV-PST-3PL**

'If they liked; they liked'

The full set of 14 SEA-like *root-V(j)VC*/*root-VCVC* dialects from Adjarian's sample is listed in Table 25. We likewise include sources on subdialects of these dialects, such that these subdialects are reported to have the same pattern.[14] The SEA pattern is the most common one, accounting for nearly half (14/31) of Adjarian's sample. It is also the conservative pattern of Standard Western Armenian.

4.5 Dialects where either type of synthetic past has been prosodically reduced

4.5.1 Subjunctive past *root-VVC*/*root-VjC*, and perfective *root-VCVC*

In SEA, the subjunctive V-V sequence /-e-i/ is pronounced with a glide [e(j)i]. But in some dialects, the V-V sequence /e-i/ has changed to a VC sequence [ej] (Hamshen, Artvin[15]) or to two identical vowels /-i-i/ (Astrakhan).[16] These three dialects retain the default past perfective template /-t͡sʰ-i/ (Table 26).

[14]Kayseri might be treated as a distinct dialect (Martirosyan 2019: 198), as can Aslanbeg (ibid., 199).

[15]For Artvin, one grammar however describes this dialect as having the SEA-pattern with the subjunctive past template /-e-i/ (Գրիգորյան 1957: 442-5).

[16]For the Astrakhan dialect, Adjarian provides some past perfectives with /-t͡sʰ-i/ and some with /-∅-ɑ/ (Աճառեան 1911: 84), though it seems that the latter template is intended for high-frequency or irregular verbs.

Table 25: Dialects or subdialects that follow the SEA pattern

Dialect	Adjarian page	Other grammars or subdialects
Akn	233	Maxudianz 1911: 87
Arapgir	215*	Դաւիթ-Բէկ 1919: 215,250; Kayseri: Անթոսյան 1961: 91
Crimea	265	New Nakhichevan: Աճառյան 1959: 416-419, Աճառեան 2021: 144
Evdokia	233	Խաչատրյան 2016: 81,83,85; Merzifon: Թումայեան 1930: 89,90
Istanbul	252	Աճառյան 1941: 120
Kharberd-Yerznka	170	Կոստանդյան 1979: 86,100,105;
		Բաղրամյան 1960: 22 (Çarsancak: 41, Çemişgezek:29)
Malatya	197*	Դանիելյան 1967: 120,122
Nicomedia	242	Գևորգյան 2016: 54
		Adapazar: Գևորգյան 2017: 101,102; Aslanbeg: Աճառեան 1898
Rodosto	260	Մեսրոպյան 2016: 100,109
Tbilisi	55,56	–
Trabzon	179	–
Sebastia	227	–
Şebinkarahisar	145*	Խաչատրյան 1985: 148,150-153
Smyrna	239*	Anonymous 1835, Vaux 2012: 33

Table 26: Dialects where the subjunctive vowel-sequence is altered

Dialect	Astrakhan	Artvin	Hamshen
Example	ing-n-**i-i̱-nkʰ**	χos-**e-j̱-nkʰ**	siɾ-**á(j)-∅̱-kʰə**
	fall-VX-TH-PST-1PL	speak-TH-PST-1PL	like-TH-PST-1PL
	'if we were falling'	'if we were speaking'	'if we were liking'
Primary source	Աճառեան 1911: 84	Ալավերդյան 1968: 238	Աճառյան 1947: 125-7
Adjarian page		292*	187

4.5.2 Subjunctive past *root-VC*, and perfective *root-VCVC*

Among dialects that reduce the subjunctive past, it is common for the /-e-i/ sequence to be reduced to a single vowel, thus creating the prosodic template *root-VC*. Unlike Iranian Armenian which also exhibits this change, many have retained their inherited default /-t͡sʰ-i/ past perfective. The old dialect of Yerevan reported by Adjarian serves as an example (Table 27).

Table 27: Old Yerevan Armenian uses past perfective template /-t͡sʰ-i/ but a single vowel in the subjunctive past for E-Class 'to like'

	Subjunctive past 3PL		Past perfective 3PL	
SEA	siɾ-**ej-i̱-n**	root-VCVC	siɾ-**e-tsʰ-i-n**	root-VCVC
Yerevan	siɾ-∅-**i̱-n**	root-VC	siɾ-**e-tsʰ-i-n**	root-VCVC
	$\sqrt{}$-**TH-PST-3PL**		$\sqrt{}$-**TH-PFV-PST-3PL**	

These dialects (Table 28) utilize mismatching prosodic templates to at least some degree.

The subjunctive past uses *root-VC*, while the past perfective uses *root-VCVC*.[17] In practice, they represent a cline. For example, Yerevan is reported to have mismatching templates: *root-VCVC* for the perfective (/-t͡sʰ-i/), and *root-VC* for the subjunctive past ([-i]). But the Lori subdialect of Yerevan has matching templates *root-VCVC* without vowel deletion: perfective /-t͡sʰ-i/ and subjunctive past [-e-i] (Ասատրյան 1968: 122,123). The Ararat subdialect of Yerevan has optional mismatch: the past perfective is *root-VCVC* (/-t͡sʰ-i/), and the vowel /e/ can optionally delete in the subjunctive past /-e-i/, creating a *root-VC* or *root-V(j)VC* template (Մարկոսյան 1989: 149-155). Some Karabakh subdialects also have matching templates *root-VCVC* (Burdur: Մկրտչյան 1971: 121,126; Goris: Մարգարյան 1975: 187,197). Though Burdur could be its own dialect (Martirosyan 2019: 232).

Table 28: Dialects or subdialects which retain past perfective template /-t͡sʰ-i/, while reducing the subjunctive past to [i] in the E-Class

Dialect	Adjarian page	Other grammars or subdialects
Julfa	89	Vaux in prep, Աճառյան 1940: §272
Karabakh	68	Դավթյան 1966: 162,170; Աճառեան 1901: 166,169
Shamakhi	79	–
Van	145	Աճառյան 1952: 159
Yerevan	42	Shamshadin: Մեժունց 1989: 80
Tigranakert	163*	Հանեյան 1978: 133, Urfa/Edessa: Ղարիբյան 1958: 155,156

4.5.3 Subjunctive past *root-VC*, and perfective *root-VCVC* or *root-VC*

Some dialects occupy an intermediate position, where they show the prosodic template *root-VC* in the subjunctive past and use both *root-VCVC* or *root-VC* among different past perfectives.

For example, in the Alashkert subdialect of Mush (Մադաթյան 1985: 121-124), the subjunctive past uses the prosodic template *root-VC* where the vowel is /e/.[18] The past perfective is optionally *root-VC* or *root-VCVC* (Table 29). Similar facts are reported for the general dialect of Mush (Կատվալյանը et al. 2016: 185) and the Bayazit subdialect of Yerevan (Կատվալյան 2016: 418-419,421). These sources likewise report the perfective template /-∅-A/ for some or all intransitive E-Class verbs in Alashkert and Bayazit. [19]

[17] Some minor segmental differences are reported. For the dialect of Karabakh, Adjarian reports the perfective template /-t͡sʰ-e/ (Աճառեան 1911: 68), while Davtyan reports /-t͡sʰ-i/ (Դավթյան 1966: 162). The Vozim subdialect of Van has the past perfective template /-t͡sʰ-ei̯/ and the subjunctive past template [-e] (Աճառեան 1911: 149). There is debate on the dialectal vs. subdialectal status of Urfa/Edessa (Martirosyan 2019: 207).

[18] Adjarian (Աճառեան 1911: 119) reports that the vowel in the past subjunctive is [i] for Mush.

[19] The Sason subdialect of Mush is reported to have the prosodic template *root-VC* with [e] for the subjunctive past, but the prosodic template *root-VCVC* with /-t͡sʰ-ə/ (Պետոյան 1954: 185,212). Martirosyan (2019) reports that Sason (p. 212) and Bayazit (p. 226) are also analyzable as separate dialects.

Table 29: Optional use of matching prosodic templates in Alashkert subdialect of Mush Armenian for E-Class 'to reap'

	Subj. Pres. 3PL	Subj. Past 3PL	Past Perfective 3PL
SEA	kʰɑʁ-**e**-**n**	kʰɑʁ-**ej**-**i̲**-**n**	kʰɑʁ-**e**-**t̲s̲ʰ̲**-**i̲**-**n**
	√-**TH**-**3PL**	√-**TH**-**P̲S̲T̲**-**3PL**	√-**TH**-**P̲F̲V̲-P̲S̲T̲**-**3PL**
	root-VC	root-VCVC	root-VCVC
Alashkert	kʰɑʁ-**i**-**n**	kʰɑʁ-**e̲**-**n**	kʰɑʁ-**e**-**t̲s̲**-**i̲**-**n**
			kʰɑʁ-∅-∅̲-**i̲**-**n**
	√-**TH**-**3PL**	√-**P̲S̲T̲.T̲H̲**-**3PL**	√-**TH**-**P̲F̲V̲-P̲S̲T̲**-**3PL**
	root-VC	root-VC	root-(VC)VC

Some dialects subject the choice between *root-VC* and *root-VCVC* to a semantic distinction in the past perfective. For example, in the Shatakh subdialect of Van (Table 30), the subjunctive past uses the prosodic template *root-VC* with [e]. Transitive E-Class verbs use the perfective template /-t͡sʰ-i/, while intransitive E-Class verbs use /-∅-æ/ (Սուրադյան 1962: 134-136).

Table 30: Choice of template based on transitivity in the Shatakh subdialect of Van Armenian

	'to untie'	'to stand'		
Subj. Pres. 2SG	ært͡sʰk-**i**-**s**	χɑngʲ-**i**-**s**	√-**TH**-**2SG**	root-VC
Subj. Past 2SG	ært͡sʰk-**e̲**-**s**	χɑngʲ-**e̲**-**s**	√-**T̲H̲.PST**-**2SG**	root-VC
Past Perfective 2SG	ært͡sʰk-**ə**-**t̲s̲ʰ̲**-**i̲**-**ɾ**	χɑngʲ-∅-∅̲-**æ̲**-**ɾ**	√-**TH**-**P̲F̲V̲-P̲S̲T̲**-**2SG**	root-(VC)VC

4.5.4 Subjunctive past *root-VC*, and perfective *root-VC*

We are only aware of one dialect which follows the Iranian Armenian pattern with the *root-VC* template for both synthetic pasts other than IA itself: the Diyadin subdialect of Van (Կատվալյանը et al. 2016: 185,194). But unlike IA, it employs the past marker /-i/ for both rather than /-A/ for the past perfective (Table 31). It seems that the subjunctive past and past perfective are homophonous, though the sources do not say this explicitly. So, while our segmentation still provides zero morphs for TH and PFV, the Diyadin innovation could be formally analyzed as a complete syncretism.[20]

Table 31: Matching prosodic templates *root-VC* with [i] in the Diyadin subdialect of Van

Subj. Pres 2SG	Subj. Past 2SG	Past Perfective 2SG
gʰəɾ-**e**-**m**	gʲəɾ-**i̲**-**ɾ**	χəm-∅-∅̲-**i̲**-**ɾ**
write-**TH**-**2SG**	write-**T̲H̲.P̲S̲T̲**-**2SG**	drink-**TH**-**P̲F̲V̲-P̲S̲T̲**-**2SG**
root-VC	root-VC	root-VC

[20] Adjarian (Աճառեան 1911) treats Diyadin as a subdialect of Van, but Martirosyan (2019: 220) reports that most subsequent work treats Diyadin as a separate dialect.

4.5.5 Subjunctive past *root-V(j)VC*, and perfective *root-VC* or *root-VCVC*

As stated in Section 4.2, colloquial Eastern Armenian basically follows the conservative pattern of Standard Eastern Armenian, but allows *root-VC* among at least certain high-frequency E-Class verbs, particularly in the 3SG. This supports our proposed account for the Elsewhere Reversal, because the innovative /-∅-A/ form failed to spread without the innovative subjunctive past. However, as Avetyan informs us in personal communication, /-∅-A/ appears to be rapidly spreading. Thus, Colloquial Eastern Armenian is a potential counterexample to our observation. It is worth noting that Adjarian reported that the dialect of Yerevan, now the capital of the Republic of Armenia, exhibited a *root-VC* subjunctive past in 1911 before the community shifted towards Standard Eastern Armenian.

Another potential counterexample is the Karin dialect (Table 32). Adjarian discusses this dialect, but he does not acknowledge its past perfective. This omission implies that this dialect patterns like SEA (Աճառեան 1911: 110). However, in a grammar dedicated to this dialect, Mkrtchyan reports that the subjunctive past uses the prosodic template *root-V(j)VC* (like SEA), while the past perfective can vary between *root-VC* and *root-VCVC*. He states that some E-Class verbs take /-t͡sʰ-i/, some take /-∅-i/, and that often a verb can shift between the two templates (Մկրտչյան 1952: 68-69,72-74).

Table 32: No reduction in the subjunctive past, but optional reduction in the past perfective for the Karin dialect

	'to look'	'to sew'		
Subj. Past 2SG	ɑʃ-**e**-**i̲**-**ɾ**	kɑɾ-**e**-**i̲**-**ɾ**	√-TH-PST-2SG	root-V(j)C
Past Perfective 2SG	ɑʃ-**e**-**tsʰ-i̲**-**ɾ**	kɑɾ-∅-∅-**i̲**-**ɾ**	√-TH-PFV-PST-2SG	root-(VC)VC

On the other hand, it is reported that some subdialects of Karin pattern instead with SEA by having the past perfective consistently use the prosodic template *root-VCVC* with the morphological template /-t͡sʰ-i/. These subdialects are Akhaltskha (Томсон 1887: 63-65) and Khodorchur (Բաղրամյան 1976: 103); though Khodorchur is sometimes treated as a separate dialect (Martirosyan 2019: 191). It would take more detailed investigation into this specific property of Karin to assess its status.

4.6 Dialects where either type of synthetic past has been lost

Finally, there are some dialects that have replaced one or both synthetic past constructions with innovative periphrastic ones. We discuss them here briefly for completeness.

4.6.1 No subjunctive past, but with past perfective *root-VCVC*

Some dialects have lost the subjunctive past, but retain the past perfective with prosodic template *root-VCVC*. One of these is the dialect of Aramo village in Syria, which Adjarian discusses in his 1911 monograph (Աճառեան 1911: 212). For the village of Aramo, Gharibyan (Ղարիբյան 1958: 40-43) reports that the subjunctive past is marked by periphrasis with a participle and auxiliary. The past perfective uses the prosodic template *root-VCVC*.

Additional sources describe some additional Syrian villages as having lost a synthetic subjunctive past, while retaining the past perfective. But in the Kessab dialect of Syria, Cholakian

(Չոլաքեան 2009: 126-127,135-136) reports that the past perfective uses the prosodic template *root-VCVC*, while the subjunctive past is lost and replaced with the morpheme sequence of the verb stem + the morph [eɾ] + finite auxiliary. The grammarian writes these forms as a single word, but we treat the auxiliary as a clitic based on similarities to other dialects in the region (Aramo, Svedia).[21]

There is significant variation across subdialects in Cilicia. For the Hadjin subdialect, Adjarian 1911 (Աճառեան 1911: 204) reports that the indicative past imperfective (built from the subjunctive past) uses the prosodic template *root-VC*, but he does not report on the past perfective. In later work, Adjarian reports that in the Zeytun and Hadjin subdialects, the subjunctive past has the template *root-VC* and the past perfective has the template *root-VCVC* (Աճառյան 2003: 226).[22]

There is scant information on other (sub)dialects of Cilicia. One source reports that the Yozgat and Sivrihisar areas of Cilicia follow the SEA pattern where the subjunctive past is *root-V(j)VC* while the past perfective is *root-VCVC* (Մկրտչյան 2006: 75,84,85,180,187). Though these may again be separate dialects (Martirosyan 2019: 196). Besides these areas, the vernacular of Nuzger is reported to have the template *root-VCVC* for the past perfective, while the synthetic subjunctive past is lost and replaced by periphrasis with the perfective converb (Բաղրամյան 1984: 46). Based on the geographic origins of this vernacular in the Gandzak/Ganja area of Azerbaijan, this vernacular is likely a subdialect of Karabakh.

4.6.2 No subjunctive past, but with past perfective *root-VC*

In Iran, the dialects of Khoy/Urmia and Maragha have lost a synthetic subjunctive past but utilize the template *root-VC* for the past perfective. Adjarian 1911 provides little information on Khoy (Աճառեան 1911: 288), but a grammar dedicated to this dialect (Ասատրյան 1962: 105-7) reports that it uses the template *root-VC* for the past perfective. The vowel is [i] for transitive verbs, and [ɑ] for intransitive verbs, and the subjunctive past is marked by adding the past particle [eɾ] after the subjunctive present. For Maragha (Աճառեան 1911: 283), there is no transitivity effect.[23] The subjunctive past is formed as in Khoy, by combining the subjunctive present with a past particle.

4.6.3 Subjunctive past *root-V(j)VC*, but no past perfective

In the Austro-Hungarian or the Artial/Artyal/Ardeal dialect, Adjarian (Աճառեան 1911: 271) reports that the indicative past imperfective uses the prosodic template *root-V(j)VC*. In such dialects, the indicative past imperfective is built from the subjunctive past by adding a prefix like *gi*. Thus, we can assume that the subjunctive past likewise uses this same template. But, Adjarian states that the synthetic past perfective is lost. To mark the simple past, the dialect

[21]The vernacular of Svedia belongs to the Syrian group (Martirosyan 2019: 178). It patterns the same as Aramo (Աճառյան 1959: 465; Աճառյան 2003: 485).

[22]One dialectological survey implies that Zeytun has the prosodic template *root-VCVC* for both synthetic pasts, but the description is unclear (Ղարիբյան 1941: 371).

[23]In a later grammar for this dialect, Adjarian reports more variation in the choice of past vowel for the past perfective (Աճառյան 1926: 234), but the facts for the subjunctive past are the same (ibid. 277). It is possible that the vowel variation is due to vowel harmony (Աճառյան 1959: 492-495).

instead uses periphrasis with a participle and inflected auxiliary.[24]

4.6.4 No subjunctive past nor past perfective

One grammar of the Meghri dialect, which is not included in Adjarian 1911, reports the past perfective is marked by combining a finite participle with a finite auxiliary (Աղայան 1954: 204-5). The subjunctive past is marked by combining the subjunctive present with a past particle.[25]

In Agulis or Zok Armenian, we find similar changes (Աճառեան 1911: 98-100). The subjunctive past is marked by combining the subjunctive present with a past particle [nel]. For the past perfective, one strategy is to combine a past participle with a finite auxiliary. Another strategy is to add agreement marking after the theme vowel without a designated past morph. This latter strategy is inherited from the Classical Armenian indicative present.[26]

4.7 Quantitative Summary

In Section 3, we proposed that Iranian Armenian's "Elsewhere Reversal" change was indirectly caused by a change to the prosodic template of subjunctive past verbs in the E-Class. This predicts that an Armenian dialect could show the innovative subjunctive past without the innovative past perfective, and indeed several such dialects do indeed exist. A classification and count of dialects showing each state is presented in Table 33.[27]

The table is organized as follows. The first section, marked "Faithful" contains dialects following a *root-VCVC/root-VCVC* prodic pattern, such as Standard Eastern Armenian with a synthetic subjunctive past in [$\sqrt{}$-**e(j)**-**i**-AGR] and past perfective in [$\sqrt{}$-**e**-**ts^h**-**i**-AGR] for the E-Class. Of the 53 varieties surveyed, almost half of them (n = 25, 47%) follow this pattern. This conservative pattern remains far and away the most dominant.

The second group, marked "Reduced," contains dialects that have altered either of the two synthetic past constructions (n = 19, 36%). Almost half of these "Reduced" dialects have reduced the subjunctive past to something following the prosodic template *root-VC* (whether optionally or always; n = 9, 17%; third and fourth rows in the "Reduced" box). Only two, Iranian Armenian and Diyadin have extended some *root-VC* past perfective pattern obligatorily. Mush, Bayazit, and Shatakh have done so under some conditions. Two more, Colloquial Eastern and Karin, show some optional distribution of *root-VC* past perfectives despite retaining the conservative subjunctive past.

The third group, "Lost," contains dialects that have lost either of the two synthetic forms (n = 9, 17%).

[24] Partial data is also provided for the Suceava subdialect of Austro-Hungarian (Աճառյան 1959: 437-441), in a manner that matches the general Austro-Hungarian dialect.

[25] The vernacular of Karchevan behaves similar to Meghri (Ղարիբյան 1941: 152,297-299). Geographically, Karchevan is located within the Meghri province, so it could be considered a subdialect of Meghri.

[26] In a later monograph grammar for this dialect, Adjarian provides the same interpretation for the subjunctive past (Աճառեան 1935: §325), but different interpretations for the past perfective (ibid., §305,321). For the verb 'to beat', a periphrastic rendering for the past perfective is [tʰάk-ɑl ə-n] where agreement is on an auxiliary. A synthetic form is [tʰάk-ɑ-n]. Adjarian is unsure if this synthetic form uses a past morph [ɑ] (due to contact with other Armenian varieties in Iran), or if this synthetic form is a reduction of the periphrastic clitic [ə-n].

[27] The distinction between dialects and subdialects remains a challenge. We set aside subdialects that pattern like their mother dialect. For example, the Aslanbeg subdialect of Nicomedia patterns like the main Nicomedia dialect so we set it aside.

Table 33: Distribution of prosodic templates for the subjunctive past and past perfective across Armenian.

	Past Perfective	Subjunctive Past	Dialects, and subdialects from other areas	
Faithful	root-VCVC	root-V(j)VC	**Standard Eastern Armenian**, Akn, Arapgir, Crimea, Evdokia, Istanbul, Kharberd-Yerznka, Malatya, Nicomedia, Rodosto, Tbilisi, Trabzon, Sebastia, Şebinkarahisar Smyrna, Standard Western Armenian, Classical Armenian	17
			Artvin (one variant), Lori (Yerevan), Burdur (Karabakh), Goris (Karabakh), Akhaltskha (Karin), Khodorchur (Karin), Yozgat (Cilicia), Sivrihisar (Cilicia)	8
Reduced	root-VCVC	root-VVC	Astrakhan	1
	root-VCVC	root-VjC	Artvin (one variant), Hamshen	2
	root-VCVC	root-VC	Julfa, Karabakh, Shamakhi, Van, Yerevan, Tigranakert	6
			Sason (Mush), Zeytun/Hadjin (Cilicia)	2
	root-VCVC	root-V(j)VC, root-VC	Ararat (Yerevan)	1
	root-VCVC, root-VC	root-VC	Mush	1
			Bayazit (Yerevan), Shatakh (Van)	2
	root-VC	root-VC	**Iranian Armenian**	1
			Diyadin (Van)	1
	root-VCVC, root-VC	root-V(j)VC	Karin	1
			Colloquial Eastern (SEA)	1
Lost	–	root-V(j)VC	Austro-Hungarian	1
	root-VCVC	–	Aramo (Syria)	1
			Kessab (Syria), Nuzger (Karabakh)	2
	root-VC	–	Khoy, Maragha	2
			Chaylu (Karabakh)	1
	–	–	Agulis, Meghri	2
Total				53

Our findings are consistent with our proposal regarding the relationship between the innovative subjunctive past and past perfective, since this proposal allows for the existence of conservative dialects like Standard Eastern Armenian, completely innovative dialects like Iranian Armenian of which there are two clear examples with innovative past perfectives, and dialects with innovative subjunctive pasts but conservative past perfectives, of which there are 9 clear unambiguous examples.

Colloquial Eastern Armenian and Karin, which retain conservative subjunctive pasts, but for which some grammars report varying distributions for optional /-∅-A/, stand out as two potential counter-examples to our proposed relationship between the changes. However, if the distribution of /-∅-A/ in these dialects is indeed limited, mainly to certain high frequency verbs, particularly in the 3SG, then these can be treated as lexical exceptions. These dialects do not pose a problem because our hypothesis deals with the extension of the innovative /-∅-A/ pattern from irregulars to the default for the large E-Class, and not the ultimate origination of exceptional /-∅-A/.

5 Modeling the "Elsewhere Reversal"

The Iranian Armenian past perfective has been subjected to analogical extension, where a minority irregular pattern /-∅-A/ has replaced the previously default/elsewhere pattern /-t͡sʰ-i/ as the new elsewhere pattern for the largest verb class in the language. Like all cases of analogical extension from an irregular minority pattern to a majority or default, this initially appears paradoxical. How could a less frequent apparently irregular pattern take over from a higher frequency apparently regular pattern?

Child language acquisition (L1A) provides an explanation for this paradox. The process by which children learn their grammars has been implicated as one of the major drivers of language change at least since Hermann Paul's Neogrammarian treatise, *Prinzipien* (Paul 1880), and has since been taken up by several distinct research traditions including generative phonology and syntax (Cournane 2017, van Gelderen 2011, Halle 1962, Kroch 2005, Lightfoot 1979), certain diachronic criticisms of generative linguistics (Baron 1977), variationist sociolinguistics (Labov 2007, 2011), and computational modeling (Hare & Elman 1995, Kodner 2020, Niyogi & Berwick 1997, Yang 2002). While debates continue surrounding exactly which types of change are the consequence of L1A, the classic parallel between analogical change and children's over-regularization (over-generalization) errors (e.g., **feeled* instead of *felt*) is particularly well-supported. Thus we adopt over-regularization during L1A as the mechanism for the innovation of the Elsewhere Reversal.

More specifically, we argue that the change in Iranian Armenian's hiatus management strategy removed the theme vowel from its subjunctive past, and that this change opened up the opportunity learners to adopt a novel generalization in the grammar regarding the past perfective, thus leading to the analogical change. Using the Tolerance Principle (TP; Yang 2016), a quantitative model of productivity learning within L1A, we calculate which Armenian morphological patterns could be learned as productive (i.e., available for (over-)generalization) or unproductive before and after the change in subjunctive past vowel hiatus. This provides a way to test the feasibility of our hypothesis empirically: if there is at least one feasible pathway over learner over-generalizations from the conservative Standard Eastern Armenian-like to innovative Iranian Armenian paradigms after the change in vowel hiatus but not before, then we have support for our hypothesis that is complementary to what can be derived from the synchronic dialect survey or from a theoretical analysis. Furthermore, the quantitative nature of this approach allows us to make additional predictions which can be cross-checked with the dialect survey. Taken together, acquisition, dialectology, and linguistic theory provide a cohesive account of the Elsewhere Reversal.

5.1 Child Language Acquisition of a Mechanism for Actuation

Our primary focus, and the focus of most work connecting child language acquisition (L1A) and change, is on the innovation and actuation grammar change rather than community-level processes like propagation. *Actuation* refers to the incipient stages of a change as it is innovated and gains an initial foothold in the population (Labov et al. 1972). In our case, understanding actuation would tell us what set off the spread of /-∅-A/ from a narrow distribution of irregulars and lexical exceptions to the entire E-Class in Iranian Armenian.

Actuation has always proven challenging to study empirically because, unlike later stages of a change, it does not leave behind a historical record. This is the challenge of the so-called the *Actuation Problem* (Weinreich et al. 1968) in sociolinguistics. By the time linguists have become aware of a change, it has already been actuated, so it is too late to observe the process of actuation directly. Nevertheless, Weinreich et al. (1968) argue that actuation is "perhaps the most basic" question in language change (p. 102). Though we can never know the exact mental state and social embedding of the specific individuals who innovated any given change, we can still reason about the mechanisms of innovation and actuation (Milroy & Milroy 1985). This is the approach taken for analogous problems across the historical sciences. No paleontologists or evolutionary biologists could tell us exactly when and under what exact pressures a particular species evolved, but the fields have worked together to explain under what circumstances species evolve, and they can often tell us roughly where and when specific evolutionary events must have occurred. This is what we aim to do in modeling historical linguistic innovations with L1A.

One of the most striking parallels between L1A and language change is that between morphological *over-generalization* (also called *over-regularization*) on one hand, and analogical change on the other. Both describe situations in which a productive pattern is innovatively generalized past its conservative scope. The patterns that are replaced are typically, but not always, less frequent than those that replace them, and high frequency items are more resistant but not immune to change. They differ in that over-generalization unfolds in the grammar of and during the development of an individual, while analogical change unfolds in the language of a community and potentially over many lifespans.

A classic example of over-generalization of productive patterns in the typical acquisition of English is the production of **feeled* or **goed* instead of expected *felt* or *went*. It should be noted that over-regularization, while still relatively rare in learner productions, is far more frequent than *over-irregularization*, where a child might say *flee*-**flaw* extending *see-saw*. Various studies have estimated the rate of over-regularizations in English, German, and Spanish verbal productions to be between 5% and 10%, while with over-irregularizations, make up well under 1% of productions (Clahsen & Rothweiler 1993, Clahsen et al. 1992, Kodner 2020, Maratsos 2000, Mayol 2007, Xu & Pinker 1995, Yang 2002). See Marcus et al. (1992, ch. 4) and Lignos & Yang (2018) for more discussion.

The frequent over-generalization of English weak verbs with the productive *-ed* pattern is strikingly parallel to the diachronic tendency in English for these verbs to analogize to synchronically irregular paradigms of strong verbs. For example, *help-helped* overtook *help-holp* in the Middle English period. Analogical leveling of this kind has proven much more common in the history of English than the extension of any of the strong verb paradigms, but that has happened as well. A formal model of language acquisition accounts for both the ability of infrequent patterns to over-generalize and analogically extend and for the discrepancy in likelihood between high- and low-frequency patterns (Ringe & Yang 2022). If over-generalization of productive patterns is the mechanism for analogical innovations, then we make the strong prediction that analogy only occurs on the basis of productive patterns. To evaluate this, we must adopt a concrete model of productivity, which we find in the form of the Tolerance Principle in the next section.

All that is needed for learner over-regularization to snowball into analogical change is for one innovating child's over-regularization to "escape" their own grammar into their speech community (i.e., be actuated) before it is outgrown. This escape should not be a common occurrence, because language would change unrealistically fast if it were. Rather, just like in biological evolution, where most consequential mutations, even beneficial ones, never make it from the individual into the population, children should grow out individual innovations during L1A before they get a chance to be passed on.

But actuation is possible for two reasons. First is the nature of early childhood sociolinguistic embedding: children are active participants in their speech communities. They acquire grammatically conditioned variation from the people around them (Labov 1989, Miller 2013, Roberts 1997, Smith et al. 2009), they receive input from other children who are not yet linguistically mature themselves (i.e., may be innovating) (Loukatou et al. 2021, Nardy et al. 2014), and they readily converge on innovative grammatical patterns from their child peers even when these differ from the language of caregivers in both monolingual and multilingual settings (Nardy et al. 2014, Rydland et al. 2014).

Second is the extreme sparsity and skew of inflected forms in morphological paradigms, including early linguistic input. Across languages, individual items follow long-tailed Zipfian frequency distributions, and so does the distribution paradigms by the proportion of possible forms attested (see Chan (2008), Lignos & Yang (2018), and Kodner (2023a) for summaries). Such distributions are common throughout language as well as non-linguistic domains (Baroni 2005, Chan 2008, Jelinek 1997, Lignos & Yang 2018, Miller 1957, Yang 2013). In a language with a paradigm the size of Armenian's, even in millions of words of speech, most verbs from the dictionary are simply not attested. Of those that are, the median *paradigm saturation*, or proportion of possible forms attested, is only about two forms per verb, while a handful of the highest frequency verbs have higher saturations.

The sparse and skewed distribution of forms in a learner's input is itself enough to give rise to some correlational trends of analogy when conceived of as the This so-called *Paradigm Cell Filling Problem* has been put forth as a driver of analogical change (Ackerman & Malouf 2013, 2015). Since lower frequency forms are more likely to have to be inferred by the learner rather than attested in the input directly, they are more likely to be subject to analogical change over time. This yields a "conserving effect" of frequency against analogical leveling (Bybee & Thompson 1997, Bybee 1985), and the resulting corresponding synchronic correlation between frequency and irregularity. Higher frequency forms have a better chance to serve as the basis for analogy and are less likely to be leveled away.

This tendency and others can be seen in various aspects of the Elsewhere Reversal: First, the fact that irregular /-∅-A/ is observed among high token frequency verbs in conservative dialects is consistent with the conserving effect of frequency against leveling. Second, in these dialects, /-∅-A/ sometimes appears among these high frequency verbs only in the 3SG. Analogy in favor of the 3SG is consistent with Watkin's Law (Watkins 1962), which is the observation that the 3SG, especially when zero-marked, is particularly likely to serve as the basis for analogy (Janse 2009). And third, since the Elsewhere Reversal imposes a shared prosodic pattern among the past forms, it can be described as imposing paradigm uniformity (Kenstowicz 1996).

But identifying the extent to which a change conforms to descriptive tendencies of analogical change tells us nothing about what caused the actuation of said change. A satisfactory solution to the Actuation Problem must tell us not only why a change may have happened when it did, but also why changes that could happen fail to happen or fail to happen earlier (Weinreich et al. 1968: p. 112). If a language's verbal paradigms could be made more uniform by undergoing Watkin's Law, for example, the question of why the change has not yet occurred always remains. Descriptive tendencies further fail to explain why some unusual analogical change managed to flout them.

The Elsewhere Reversal's striking feature, the analogical extension of a clearly irregular, low type frequency, and apparently unproductive pattern in favor of the overwhelmingly dominant default, is such a flouting change. We contend that a concrete model of productivity learning as a mechanism for actuation can provide an explanation for the change, which a post hoc cataloging of tendencies cannot. The Tolerance Principle, which we describe in the next section, provides such a model. With a precise definition of productivity and irregularity and a quantitative model for learning productivity from data, we can make sense of the Elsewhere Reversal at a level beyond correlations and make a proposal for the actuation of the change.

5.2 The Tolerance Principle

The *Tolerance Principle* (TP; Yang 2005, 2016) is a concrete model of productive generalization learning, which is one of the core components of L1A. The TP was originally developed with morphology in mind, but it is independent of any particular level of the grammar. This separation of the algorithmic aspects of acquisition from the substance of the representations over which generalizations are formed (Payne & Yang 2023) affords it wide applicability to L1A, where it has been applied to a range of generalization learning problems in phonology (e.g., Belth 2024, Dresher & Lahiri 2022, Richter 2021), morphology (e.g., Belth et al. 2021, Björnsdóttir 2021, Henke 2023, Yang 2016), and syntax and semantics (e.g., Irani 2019, Nowenstein et al. 2020, Yang 2016). It is motivated by research on pattern learning in psycholinguistics and has been validated over a range of experimental studies, also spanning morpho-phonology to syntax and semantics (Emond & Shi 2020, Koulaguina & Shi 2019, Li & Schuler 2023, Schuler 2017, Shi & Emond 2023). It already has a strong track record in modeling language change across phonology (Dresher & Lahiri 2022, Kodner & Richter 2020, Richter 2021, Sneller et al. 2019), syntax (Nowenstein et al. 2020, Trips & Rainsford 2022), and also analogical change in morphology (Kodner 2023b, Ringe & Yang 2022). Importantly, while the TP is highly applicable to language change, it is not a model of language change *per se*. Its diachronic successes come from its success as a quantitative model of generalization learning and productivity. In other words, it was not designed *post hoc* by historical linguists to fit the data of a particular problem, so it provides field-external validation for historical hypotheses.

The TP is a type-based model which views generalizations in terms of productivity in the face of exceptions. Productivity, as conceived of for the TP, differs from other views on productivity in two key ways. First, productivity is a cognitive notion, not a corpus one. It is a property of each individual's grammar. Corpora are useful as proxies for input during L1A over which productivity can be calculated but are not the object of study in themselves. Second, productivity is a categorical concept, not a gradient likelihood of use or acceptability. Either a generalization is productive (i.e., it can be extended by a speaker to new types by rule or equivalent notion) or it is unproductive (i.e., it is memorized as lexical exceptions.).

This binary distinction has been conceptualized into many theories of grammar in terms of Elsewhere Conditions (e.g., Aronoff 1976, Kiparsky 1973) in which irregulars have to be looked up in some way before defaulting to a regular pattern. These may be global or conditioned on additional phonological or semantic information. In the Armenian case, for example, there may be multiple rules conditioned on the presence of different theme vowels (e.g., a separate default pattern for A-Class verbs that applies ahead of the Elsewhere Condition) or other finer-grained distinctions within the conjugation classes that are defined semantically or conjugationally, such as a rule that applies to specifically to intransitive verbs or to words with disyllabic stems ahead in lieu of the Elsewhere Condition.

In a system that makes a binary distinction between productive regulars and unproductive irregulars, the regulars have to be listed somehow and checked against before a regular rule or pattern can be applied. The TP provides the threshold below which it is cognitively more efficient to learn the pattern and memorize the exceptions and above which it is more efficient to memorize them all or learn a different rule. The full cognitive motivation and mathematical derivation for the TP are out of scope for this paper, but see Yang (2016, pp. 10, 144), and Yang (2021) for more information.

The formal definition is as follows. The *tolerance threshold* θ_N is the number of exceptions below which a learner should acquire a rule or generalization. θ_N is a function of N, the number of lexical items in the scope of the generalization (e.g., the number of verb roots an English learner has acquired so far), and e is the number of exceptions known to the learner so far (e.g., the number of verbs that the learner knows do not take *-ed* in the past tense). This is provided as a formula in (5). It is defined as the number of known types that a generalization should apply to divided by the natural logarithm of the number of types.

(5) **Tolerance Principle**:
If R is a productive rule applicable to N candidates, then the following relation holds between N and e, the number of exceptions that could but do not follow R:

$$e \leq \theta_N \text{ where } \theta_N := \frac{N}{\ln N}$$

To take English past inflection as a simple example, the TP says that if a learner knows $N = 500$ verbs, they will acquire the add-*ed* rule to form the past tense as long as no more than 80 ($\theta_N = 80.46$) of those verbs are known to be irregular by the child.

One important property of the TP is that learners will adjust their decisions about productivity as they learn new vocabulary and their tolerance thresholds shift accordingly. A generalization that is not productive early in a learner's development can become productive and vice-versa as that learner's vocabulary grows. This bears out in the acquisition of English past tense. Early learners' verbal lexicons are disproportionately irregular due to the clustering of irregular verbs at the high end of the frequency range. They only begin producing *-ed* productively (in observation and when prompted in Wug tests) when they have learned enough regular verbs to push θ_N above the number of exceptions in their vocabularies (Marcus et al. 1992, Yang 2016).

This falling in and out of productivity accounts for the observed asymmetry between over-generalization (over-regularization) and over-irregularization. Over-regularization is the application of a productive pattern past where it would apply in adult speech, perhaps because the learner has (temporarily) hypothesized a scope that is too broad for that pattern or because they have yet to memorize a lexical exception. On the other hand, over-irregularization would be the over-application of an unproductive pattern. But unproductive patterns are memorized, not represented as rules or equivalents, so they cannot be over-applied by the grammar.

An important consequence of this for analogical change is that all analogy is the over-generalization of *productive* patterns, even productive patterns that are infrequent or of narrow scope. For our purposes, this means that the /-∅-A/ past perfective must have been productive over some narrow scope in the ancestor of Iranian Armenian for it to have been over-generalized. If this scope only became available after the change to the subjunctive past, then it could not have been over-generalized in other dialects and therefore could not have been analogically extended.

5.3 Modeling learners without child-directed data

Since the Tolerance Principle is a quantitative cognitive model that applies over a learner's lexicon, we need a lexicon of Armenian to work with. Ideally, we would estimate this from a corpus of child-directed speech (CDS) as is typical in computational modeling of child language acquisition. Unfortunately, there are no corpora of Armenian CDS, just as there no CDS corpora for the vast majority of the world's languages, historical and modern. While this poses a challenge to anyone attempting to quantitatively model acquisition in the past, it is nevertheless often reasonable to estimate properties of the input from non-CDS corpora.

There are three properties of acquisition and early linguistic input that facilitate corpus investigation of morphological acquisition. First, it is type frequency, not token frequency, over which a pattern is expressed that drives productivity learning. Results from a wide range of often diametrically opposed research programs agree on this point (Aronoff 1976, Baayen 1993, Bybee 1985, MacWhinney 1978, Pierrehumbert 2003, Yang 2016). Second, token frequency is nevertheless indirectly relevant for acquisition because it correlates well with the age at which an item is acquired (Goodman et al. 2008). This makes sense because high frequency items are more likely to be attested early since they have more opportunities to be so. Third, early learner vocabularies are quite small. Across languages that have been investigated, the typical learner knows only a few hundred to a thousand lemma by around age three when most of morphology acquisition has taken place (Fenson et al. 1994, Hart & Risley 1995, 2003, Kern 2003, Szagun et al. 2006) and only a third to half of those are verbs (Bornstein et al. 2004).

Given these three properties, the few hundred most frequent types in a corpus of CDS makes a good vocabulary estimate for a child learner (Nagy & Anderson 1984, Yang 2016). Kodner (2019) demonstrates that historical and other non-child-directed corpora can be substituted for CDS when the same processing steps are applied. While lexicons derived from CDS and non-

CDS corpora do differ, the high frequency items from these corpora are roughly as similar as those derived from different CDS corpora are to one another. We take inspiration from their approach in applying the Tolerance Principle at several different vocabulary sizes in order to model learners developmental trajectories. Calculating productivity on the top two hundred most frequent verbs from a corpus should roughly model a three-year-old, while considering the top five hundred verbs would model a child closer to age six or seven.

5.4 Methods

Our proposed pathway for the Elsewhere Reversal requires that some learner first innovate the Iranian Armenian grammar where the past tense is expressed as *root-VC*, with surface /-i-/ indicating the subjunctive past and surface /-A-/ the indicative past perfective. The pathway would still apply even if one adopts an alternative synchronic account for Iranian Armenian in which there is no longer an underlying /-e-/ theme vowel in E-Class synthetic past forms. Once any learner has hypothesized such a grammar, it is up to them to evaluate it against their input. A child learning a morphophonologically conservative variety like Standard Eastern Armenian would be forced to reject this quite quickly, because every single subjunctive past form would count against the generalization. In contrast, a learner receiving input from a variety with the innovative subjunctive past would have several opportunities to extend it, as every subjunctive past form as well as some past perfective forms support it. Other children receiving novel productions from the innovating child would not need to re-actuate them since they are now evidenced directly in their input. With this additional evidence, these subsequent children could extend the generalization even further, pass it on, and so on, until it reached the Iranian Armenian pattern. The initial evidence to the learner for this generalization is summarized in (6).

(6) Initial evidence for a learner hypothesizing the following generalization: "*The synthetic past matches the root-VC prosodic template (and the suffix vowel indicates aspect)*"

SEA-like before subjunctive past innovation: Every subjunctive past stem is an exception to the hypothesized generalization, and all past perfective verbs except for the mostly nasal-infixing and suppletive verbs with /-∅-A/ serve as exceptions to the hypothesis. It would stand no chance if it was even hypothesized by some learner in the first place.

Pre-IA after subjunctive past innovation: The hypothesis now holds for every subjunctive past stem. It still does not hold for most past perfective stems except for those that already permit the /-∅-A/ past perfective and perhaps high-frequency verbs that optionally permit it. A child might hypothesize this generalization for (some subset of) irregular E-Class verbs. This provides the analogical extension with a tenuous initial foothold.

From this point on, it is an empirical question whether the innovative generalization could ever be over-generalized given the lexical makeup of Armenian. Of course, keeping the Actuation Problem in mind, we cannot know exactly what happened on that day that the Elsewhere Reversal was actuated in Iran, but we can approach this moment asymptotically with our quantitative approach. We test many logically possible pathways for the actuation of the change.

Beginning with the verbs which conservatively form /∅-A/ past perfectives as the initial evidence in favor of the innovative grammar, we ask to which subsets of the verbal lexicon the first child should be able to productively extend /∅-A/ to given their input.

Having established that Standard Eastern Armenian fairly represents the conservative state for the relevant phenomena in Section (4.3), we use type and token counts sourced from the Ghazaryan frequency dictionary of Standard Eastern Armenian (Ղազարյան 1982), to estimate early learner vocabularies in the pre-Elsewhere Reversal Iranian Armenian. In order to confirm that our results are not dependent on the specific makeup of the Ghazaryan dictionary, we perform the same calculation on a second data set collected from the Armenian corpora of the Universal Dependencies treebank (UD; Zeman et al. 2021). Only the most frequent verbs were extracted from each corpus in order to better approximate the vocabulary of a typical learner, following Kodner (2019). Since the highest frequency items tend to be conserved across corpora regardless of genre, we should expect the calculations over Ghazaryan and UD to be similar. To model changes in productivity as a learner's vocabulary grows, we perform calculates on the top V most frequent verb lemmas at intervals $V = 50, 60, 70, 80, 90$ and $V = 100, 150, 200, \ldots, 600$.[28] The verbs were annotated to indicate what kind of inflections they take in conservative varieties, which conjugational class they belong to, and all other factors discussed in the previous sections. N is $2V$, since the calculation is performed over the subjunctive past and past perfective stems both of which should adopt the *root-VC* pattern.

With this approach, we can pose questions of the form, "Can a child who has adopted the innovative generalization and only received pre-modern Iranian Armenian input extend /∅-A/ to all E-class intransitive verbs?" (The answer is affirmative for this particular question up to V=100 on both data sets). We can then ask if a learner who received input including that generalization could extend the generalization even farther along. This calculation is repeated iteratively to model successive cohorts of learners further incrementing the grammar towards the attested states in Iranian Armenian.

We call these the *feasible actuation pathways* for the change. Nobody can say for sure which of these feasible pathways was actually taken, but this approach can eliminate some logically possible pathways that are shown to be infeasible. Furthermore discovering at least one feasible pathway provides a demonstration that our proposal for the Elsewhere Reversal is possible. If any of the feasible pathways contains an intermediate state that is attested in a modern dialect, then this would provide additional confirmatory evidence. Conversely, if no feasible actuation pathway is found, then there is some flaw in our work, either our proposal for the Elsewhere Reversal or choice of historical actuation model.

When considering possible learner generalizations, we identify the factors in (7) based on prior literature and our own observations of the data, all of which are described in Sections 2 and 4.3. We also observe that the verbs which optionally allow /-∅-A/ in Colloquial Eastern Armenian (§4.2) tend to have disyllabic stems, as do most if not all irregular verbs, so we include this possible generalization as well.

(7) Possible dimensions of generalization for our model learners

Conjugation class: A-Class vs. E-Class ((/∅-A/ is never productive for any subset of the A-Class, so we omit these calculations from the report to save space)

[28] We found that productivity calculations did not change past $V = 600$, so we describe this as *All*

Irregularity: suppletive and/or nasal-infixing vs. neither (This generalization is consistently very robust, since exceptions to /∅-A/ among this set are very limited (§2.5). so we omit these calculations as well)

Transitivity: transitive vs. intransitive

Derivational status: simplex vs. complex

Length: disyllabic stem ($\sqrt{}$-TH) vs. longer stems

Our model crucially does not require that a child maintain an innovative over-generalization into adulthood, but rather just long enough to pass it on to a new child who could then either learn it faithfully or perhaps extend it further. However, this still requires that a child hold onto the over-generalization at least long enough to transmit it. The verbal vocabulary size V in our calculations can be viewed as a very rough proxy for age, so an innovation retained up until a higher V corresponds to retention to a higher age. We find that $V = 100$ is the maximum cutoff that would allow for at least one feasible actuation pathway using our data sets. While no estimates of Armenian learners' lexicon composition exist, we know from estimates of several other languages (Bornstein et al. 2004), that verbs make up between one third and one half of a child's lexicon. Thus V=100 corresponds to the lexicons of English and German learners perhaps about three years old (Fenson et al. 1994, Szagun et al. 2006). However, this probably underestimates ages in the Iranian Armenian case, since native multilingual learners, such as the Armenian-Persian multilingual community in Tehran, tend to have smaller vocabulary sizes in each language at the same age as compared to monolingual learners (Cobo-Lewis et al. 2002, Doyle et al. 1978). This provides an indirect means by which multilingualism may have facilitated the change to Iranian Armenian. In slowing down the rate of learners' vocabulary growth, multilingualism would a longer period of peer-to-peer transmission of innovations before learners adopt adult-like generalizations.

Our final modeling assumption is that none of the past perfectives that optionally exhibit /-∅-A/ in Colloquial Eastern Armenian count in favor of our calculations. This is a conservative assumption because it raises the number of exceptions e for each of our TP calculations. In other words, it makes it less likely that the TP calculations will turn up feasible actuation pathways. In making this assumption, we uncover the minimum number of feasible acquisition pathways for the reported V cutoff. The true number may be higher.

5.5 Results

Figure 1 summarizes the feasible actuation pathways available to model Armenian learners according to our analysis, calculated over both the Ghazaryan and UD data sets. The left-hand side represents the conservative state (an SEA-like pre-modern Iranian Armenian dialect but with the innovative subjunctive past), and the right-hand side represents the attested Iranian Armenian state of the Elsewhere Reversal, where /∅-A/ has become the default for the entire E-Class. Blocks in the middle correspond to intermediate states where /∅-A/ is applied productively to subsets of the E-Class according to (7). Arrows represent paths between the states, and the numbers indicate the maximum V at which the new generalization can be made from the previous state.

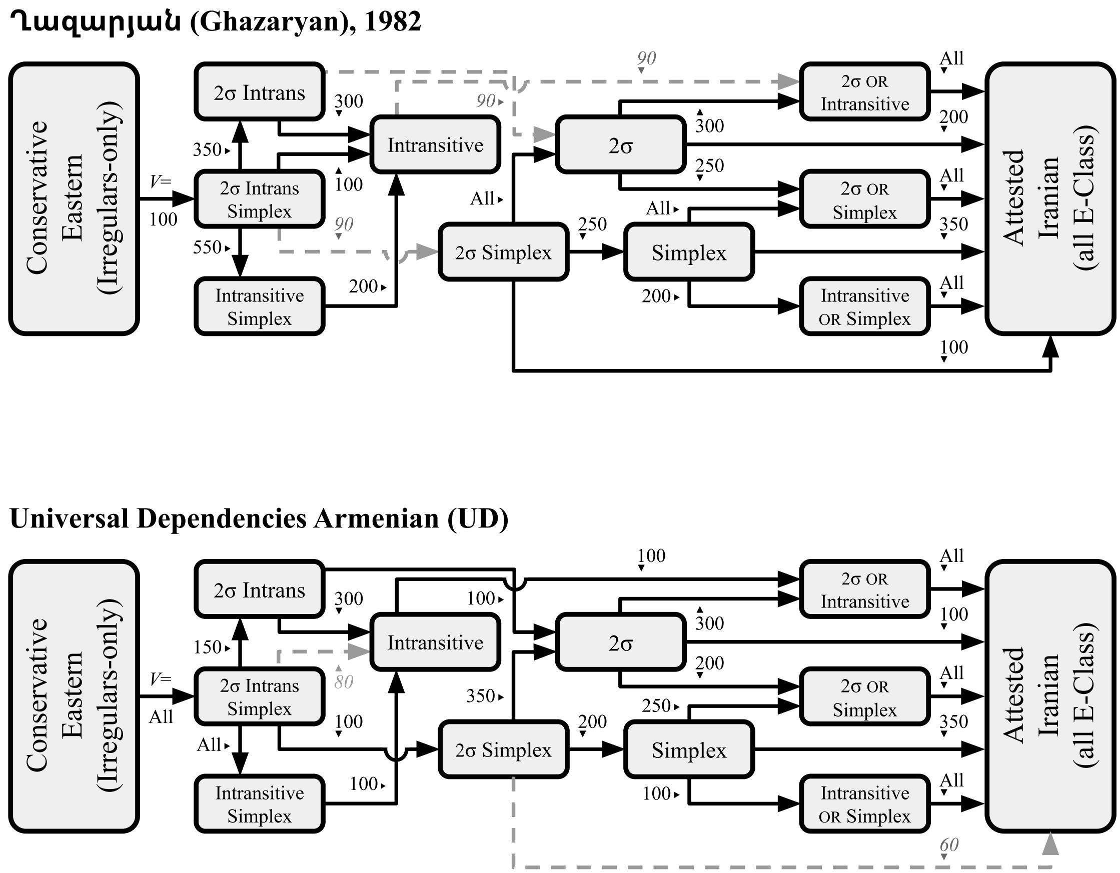

Figure 1: Summary of feasible pathways for default past perfective /∅-A/ with productive over-generalizations on both data sets. 2σ indicates a generalization to disyllabic stems. Numbers indicate the maximum V for which an over-generalization is tolerable. Solid lines indicate paths reachable if we assume the minimum V necessary for transmission is 100. Dashed lines indicate generalizations that are tolerable at at least $V = 100$ only when calculated over the alternative data set.

The feasible actuation pathways derived from both data sets are nearly identical. There are, however, five paths that met our self-imposed $V = 100$ minimum threshold on one data set but fall just below it on the other. These are represented as grey dashed lines. This is the small but real degree of variation that we can expect when taking estimates from different corpora just as we can expect variation between the input of individual children.

Tables 34 and 35 summarize the paths represented in the figures. Productivity was calculated using the Tolerance Principle, where N counts the subjunctive past and past perfective of each verb that meets the generalization among the top V most frequent verbs. Bold cells in the tables indicate productivity. Italicized cells indicate what we call "marginal" productivity: e landed within 1 of θ, which raises the possibility that the count is dependent on the particular corpus or that individuals who receive only slightly different input may vary in their generalizations.

It is clear from these results that all three potential dimensions of generalization are necessary for the /-∅-A/ to "claw its way up" from its initially narrow distribution gradually to

Input	**→Over-Gen**	. . . V=100	V=150	V=200	V=250	V=300	V=350. . .	V=550	V=All
(Init)	→ 2IS	*17 (10)*	23 (14)	31 (22)	35 (26)	37 (27)	43 (33)	66 (55)	101 (88)
2	→ (IA)	**72 (22)**	**116 (35)**	**161 (54)**	203 (74)	243 (98)	287 (126)	462 (227)	1595 (1212)
	2 ∪ S	**63 (17)**	**94 (29)**	**130 (43)**	***159 (55)***	188 (73)	219 (91)	329 (153)	1008 (717)
	2 ∪ I	**28 (9)**	**45 (10)**	**64 (17)**	**84 (28)**	**97 (34)**	118 (46)	212 (93)	786 (598)
I	→ 2 ∪ I	(See auxiliary table below)							
S	→ (IA)	**72 (9)**	**116 (22)**	**161 (31)**	**203 (44)**	**243 (55)**	**287 (68)**	**462 (133)**	1595 (587)
	2 ∪ S	**49 (4)**	**80 (16)**	**106 (20)**	**128 (25)**	**144 (30)**	**160 (33)**	**234 (59)**	**380 (92)**
	I ∪ S	**28 (6)**	**45 (16)**	**64 (24)**	84 (36)	97 (45)	118 (55)	212 (106)	786 (462)
2I	→ 2	(See auxiliary table below)							
	I	**28 (9)**	**45 (10)**	**64 (17)**	**84 (28)**	**97 (34)**	118 (46)	212 (93)	786 (598)
2S	→ (IA)	**72 (26)**	116 (51)	161 (74)	203 (99)	243 (128)	287 (159)	462 (286)	1595 (1304)
	2	**49 (4)**	**80 (16)**	**106 (20)**	**128 (25)**	**144 (30)**	**160 (33)**	**234 (59)**	**380 (92)**
	S	**63 (17)**	**94 (29)**	**130 (43)**	***159 (55)***	188 (73)	219 (91)	329 (153)	1008 (717)
IS	→ I	**28 (6)**	**45 (16)**	**64 (24)**	84 (36)	97 (45)	118 (55)	212 (106)	786 (462)
2IS	→ I	**28 (11)**	45 (22)	64 (33)	84 (49)	97 (60)	118 (75)	212 (146)	786 (684)
	2I	**19 (2)**	**35 (12)**	**47 (16)**	**56 (21)**	***63 (26)***	***72 (29)***	119 (53)	187 (86)
	2S	(See auxiliary table below)							
	IS	**22 (5)**	**29 (6)**	**40 (9)**	**48 (13)**	**52 (15)**	**63 (20)**	***106 (40)***	324 (222)
2 ∪ I	→ (IA)	**72 (13)**	**116 (25)**	**161 (37)**	**203 (46)**	**243 (64)**	**287 (80)**	**462 (134)**	1595 (614)
2 ∪ S	→ (IA)	**72 (5)**	**116 (6)**	**161 (11)**	**203 (19)**	**243 (25)**	**287 (35)**	**462 (74)**	1595 (495)
I ∪ S	→ (IA)	**72 (3)**	**116 (6)**	**161 (7)**	**203 (8)**	**243 (10)**	**287 (13)**	**462 (27)**	**1595 (125)**

Input	**→Over-Gen**	. . . V=60	V=70	V=80	V=90	V=100	. . .
I	→ 2 ∪ I	***32 (15)***	**36 (15)**	***42 (18)***	***46 (20)***	49 (23)	
2I	→ 2	***32 (15)***	**36 (15)**	***42 (18)***	***46 (20)***	49 (23)	
2IS	→ 2S	**30 (13)**	**34 (13)**	**39 (16)**	***42 (18)***	45 (21)	

Table 34: Tolerance Principle calculations over the Ghazaryan dataset for the paths reachable with $V = 100$ as a minimum for transmission. Each row corresponds to a solid path in Figure 1. Numbers indicate the number of relevant verbs for the row's generalization at each V and in parentheses, the number of exceptions to that generalization. The small auxiliary table provides smaller V for paths that are feasible at $V = 100$ for UD but not for the Ghazaryan dictionary. Bold cells indicate productivity, and italicized cells indicate marginal productivity (e is less than one away from either side of the threshold). These tables only contains V relevant to the figure. 2 = disyllabic, I = intransitive, S = simplex

the Elsewhere Reversal, as it is necessary to pass through a state that is conditioned on each dimension. For example, the generalization could expand from the conservative system after the morphophonological change to the subjunctive past, extending /-∅-A/ to all disyllabic intransitive simplex verbs (2σ Intrans Simplex), to all intransitive verbs (Intransitive), to all verbs that are either disyllabic or intransitive (2σ OR Intransitive), to the attested IA system. Other paths could also take it through disyllabic simplex verbs (2σ Simplex) and either all disyllabic verbs (2σ) or all simplex verbs (Simplex).

The Intransitive state (i.e., extension of /-∅-A/ to all intransitive E-Class verbs) appears to be a bottleneck, because while it is possible to transition past it with a $V = 100$ threshold according to the UD data set, the same calculation on the Ghazaryan data set is only tolerable up to $V = 90$. A population whose dialect made this innovation might find itself stuck in this state, maybe for a long while. If transitioning out from this state is particularly unlikely, we might predict that some attested dialects indeed got "stuck" here. This prediction is indeed born out in several dialects recorded in the descriptive grammars. These include the Shatakh subdialect of Van (closely related to the Diyadin subdialect of Van that patterns like IA), the Bayazit subdialect of Yerevan, and according to some sources, the Alashkert subdialect of Mush (Section 4.5.3). The existence of such "stuck" transitional dialects predicted by our model of Iranian Armenian is promising, and future descriptive work on Armenian which looks for these patterns specifically may uncover more dialects that fall into predicted states.

6 Conclusion

We investigate an instance of Elsewhere Reversal in Iranian Armenian verbs, in which an apparently default pattern in conservative varieties was restricted to a narrow distribution while a formerly irregular pattern became the new default. This is the consequence of a dramatic analogical extension which spread the past perfective pattern of suppletive and nasal suffixing verbs to the default E-Class, an extension from about 40 verbs to about 80% of all verbs in the language.

We approach an explanation for the Elsewhere Reversal from three complementary directions. First, we provide a theoretical analysis which gives the change its name. We see this as providing an implementation for the change in the grammar, which helps ground its ultimate distribution in the language. Second, we present a dialectological survey of inflectional patterns in Armenian varieties. This approach allows us to observe the distribution of Elsewhere Reversal-like changes and related changes synchronically in lieu of non-existent diachronic records of Armenian dialects. We observe that the two synthetic past constructions generally agree in their prosodic template. Of those varieties that do not, nearly all show the reduced template in the subjunctive past and not the past perfective. This suggests that the change to the subjunctive past plays a facilitatory or catalyst role in the change to the past perfective.

Third, we apply a quantitative model of generalization learning to explain the actuation of the analogical change. This allow us to model the mechanisms of processes of change and make precise predictions about diachronic pathways in a way that complements the theoretical model of the change's implementation and the dialectological evidence showing its correlates with other parts of the paradigm. We find that our proposal relying on the change to the subjunctive past to facilitate the Elsewhere Reversal is indeed feasible quantitatively. Our approach further

Input	**→Over-Gen**	V**=100**	V**=150**	V**=200**	V**=250**	V**=300**	V**=350**	V**=550**	V**=All**
(Init)	→ 2IS	***11 (7)***	14 (10)	18 (14)	25 (21)	31 (27)	37 (32)	55 (50)	82 (77)
2	→ **(IA)**	**76 (28)**	118 (46)	165 (63)	210 (78)	254 (97)	292 (120)	469 (193)	772 (319)
	2 ∪ S	**58 (22)**	***85 (33)***	***119 (44)***	147 (55)	177 (67)	200 (81)	317 (131)	505 (213)
	2 ∪ I	**33 (10)**	**46 (17)**	**66 (26)**	**91 (32)**	***111 (42)***	134 (54)	213 (86)	344 (136)
I	→ 2 ∪ I	***40 (18)***	64 (36)	94 (55)	122 (65)	145 (80)	160 (84)	262 (141)	438 (237)
S	→ **(IA)**	**76 (18)**	**118 (33)**	**165 (46)**	**210 (63)**	**254 (77)**	***292 (92)***	469 (152)	772 (267)
	2 ∪ S	**40 (12)**	**64 (20)**	**94 (27)**	**122 (40)**	**145 (47)**	**160 (53)**	262 (90)	438 (161)
	I ∪ S	**33 (13)**	46 (23)	66 (36)	91 (50)	111 (59)	134 (72)	213 (119)	344 (204)
2I	→ 2	***40 (18)***	64 (36)	94 (55)	122 (65)	145 (80)	160 (84)	262 (141)	438 (237)
	I	**33 (10)**	**46 (17)**	**66 (26)**	**91 (32)**	***111 (42)***	134 (54)	213 (86)	344 (136)
2S	→ **(IA)**	(See auxiliary table below)							
	2	**40 (12)**	**64 (20)**	**94 (27)**	**122 (40)**	**145 (47)**	**160 (53)**	262 (90)	438 (161)
	S	**58 (22)**	***85 (33)***	***119 (44)***	147 (55)	177 (67)	200 (81)	317 (131)	505 (213)
IS	→ I	**33 (13)**	46 (23)	66 (36)	91 (50)	111 (59)	134 (72)	213 (119)	344 (204)
2IS	→ I	(See auxiliary table below)							
	2I	**19 (8)**	**25 (11)**	36 (18)	54 (29)	62 (31)	73 (36)	118 (63)	198 (116)
	2S	***28 (14)***	44 (27)	67 (46)	82 (54)	98 (64)	107 (67)	172 (114)	277 (192)
	IS	**20 (5)**	**23 (5)**	**30 (8)**	**41 (11)**	**52 (14)**	**62 (18)**	**94 (30)**	**140 (48)**
2 ∪ I	→ **(IA)**	**76 (18)**	**118 (29)**	**165 (37)**	**210 (46)**	**254 (55)**	**292 (66)**	**469 (107)**	**772 (183)**
2 ∪ S	→ **(IA)**	**76 (6)**	**118 (13)**	**165 (19)**	**210 (23)**	**254 (30)**	**292 (39)**	**469 (62)**	**772 (106)**
I ∪ S	→ **(IA)**	**76 (5)**	**118 (10)**	**165 (10)**	**210 (13)**	**254 (18)**	**292 (20)**	**469 (33)**	**772 (63)**

Input	**→Over-Gen**	. . . V**=60**	V**=70**	V**=80**	V**=90**	V**=100**	. . .
2S	→ **(IA)**	**42 (17)**	52 (25)	59 (29)	67 (35)	76 (40)	
2IS	→ I	**15 (5)**	***21 (11)***	***26 (13)***	28 (15)	33 (18)	

Table 35: Tolerance Principle calculations over UD for the paths reachable with $V = 100$ as a minimum for transmission. Each row corresponds to a solid path in Figure 1. Numbers indicate the number of relevant verbs for the row's generalization at each V and in parentheses, the number of exceptions to that generalization. The small auxiliary table provides smaller V for paths that are feasible at $V = 100$ for the Ghazaryan data but not for UD. Bold cells indicate productivity, and italicized cells indicate marginal productivity (e is less than one away from either side of the threshold). These tables only contains V relevant to the figure. 2 = disyllabic, I = intransitive, S = simplex

makes positive predictions regarding the existence of Armenian varieties that have only extended the conservative pattern as far as intransitive E-Class verbs, which is clearly borne out in the dialectological survey. While nobody can ever hope to observe the moment that a change is actuated, such a concrete mechanism brings us closer to answering the "why here and now?" and "why not earlier?" sides of the actuation question. These insights could not have been drawn from an account that relies on descriptive tendencies like Watkin's Law remaking paradigms on the basis of the 3SG, a tendency toward leveling in favor of more frequent forms, or a general drive toward paradigm uniformity, as these cannot tell us the specific conditions that catalyze a change, nor can they answer why a given preferred change did not simply occur earlier in the history of a language. To put it another way, if there is a drive towards paradigm uniformity, why has it not already occurred in the observed initial state of the language? Only with a concrete mechanism in hand can we state that a prior change was prerequisite.

Taken together, we have provided an account for the Elsewhere Reversal, casting it as a normal, albeit unlikely, case of analogical leveling, which is itself an occasional long-term consequence of normal language acquisition. When a novel productive generalization becomes available in the lexicons of early learners (perhaps triggered by an unrelated phonological change), there is a chance for children to extend it through over-generalization. If the right words have high enough token frequency to be acquired early, and the type frequencies of competing patterns in the lexicon happen to be just right, the constrained over-generalization can continue to expand. We expect that such a perfect alignment of the stars should be quite rare compared to more typical leveling of frequent dominant patterns but not impossible (see Ringe & Yang (2022) and Kodner (2023b) for related treatments in English and Latin). By approaching this problem from three directions, we have been able to provide both an implementation and mechanism as well as a quantitative cross-dialectal validation of our proposal. Thus analogical extensions like the Elsewhere Reversal may be explained as mere leveling against the odds under identifiable conditions.

Abbreviations

- AGR: agreement
- ASP: aspect
- AUX: auxiliary
- CAUS: causative
- CEA: Colloquial Eastern Armenian
- DEF: definite
- IA: Iranian Armenian
- IMP: imperative
- IMPF.CVB: imperfective converb
- INCH: inchoative

- IND: indicative
- INF: infinitive
- LV: linking vowel
- PASS: passive
- PERF.CVB: perfective converb
- PFV: perfective
- PL: plural
- PST: past
- RPTCP: resultative participle
- SG: singular
- SEA: Standard Eastern Armenian
- SWA: Standard Western Armenian
- T: tense
- TH: theme vowel
- VX: verb extender

References

Ackerman, Farrell & Robert Malouf. 2013. Morphological organization: The low conditional entropy conjecture. *Language* 89(3). 429–464.

Ackerman, Farrell & Robert Malouf. 2015. The No Blur Principle effects as an emergent property of language systems. In *Proceedings of the annual meeting of the Berkeley Linguistics Society*, vol. 41 41, 1–14.

Adjarian, Hrachia. 1909. *Classification des dialectes arméniens*. Paris: Librairie Honoré Champion. `https://archive.org/details/bibliothquedel173ecol/page/n7/mode/2up`.

Anderson, Stephen R. 1992. *A-morphous morphology*. Cambridge: Cambridge University Press 1st edn. doi:10.1017/CBO9780511586262.

Anonymous. 1835. *Քերականութիւն Անգղիարէն Եւ Հայերէն*. Յիզմիր: Ի տպարանի Ամերիգացւոց.

Aronoff, Mark. 1976. *Word formation in generative grammar* (Linguistic Inquiry Monographs 1). Cambridge, MA: The MIT Press.

Baayen, Harald. 1993. On frequency, transparency and productivity. In *Yearbook of morphology 1992*, 181–208. Dordrecht: Springer. doi:10.1007/978-94-017-3710-4_7.

Balabanian, George. 2024. *A diachronic analysis of western armenian verbal morphology*: University of Pennsylvania dissertation.

Baron, Naomi S. 1977. *Language acquisition and historical change*. Amsterdam: North-Holland Publishing Company.

Baroni, Marco. 2005. 39 distributions in text. *Corpus Linguistics: An International Handbook Volume* 2. 803–822.

Barry, James. 2018. *Armenian Christians in Iran: Ethnicity, religion, and identity in the Islamic Republic*. Cambridge: Cambridge University Press 1st edn. doi:10.1017/9781108684873.

Belth, Caleb. 2024. A learning-based account of phonological tiers. *Linguistic Inquiry* 1–63.

Belth, Caleb A, Sarah RB Payne, Deniz Beser, Jordan Kodner & Charles Yang. 2021. The greedy and recursive search for morphological productivity. In *Proceedings of the annual meeting of the cognitive science society*, vol. 43 43, .

Bezrukov, Nikita. 2022. *Caucasus in motion: Dynamic wordhood and morpheme positioning in Armenian and beyond*: University of Pennsylvania dissertation.

Björnsdóttir, Sigríður Mjöll. 2021. Productivity and the acquisition of gender. *Journal of Child Language* 48(6). 1209–1234.

Blevins, James P. 2006. Word-based morphology. *Journal of Linguistics* 42(3). 531–573. doi: 10.1017/S0022226706004191.

Bornstein, Marc H, Linda R Cote, Sharone Maital, Kathleen Painter, Sung-Yun Park, Liliana Pascual, Marie-Germaine Pêcheux, Josette Ruel, Paola Venuti & Andre Vyt. 2004. Cross-linguistic analysis of vocabulary in young children: Spanish, dutch, french, hebrew, italian, korean, and american english. *Child development* 75(4). 1115–1139.

Boyacioglu, Nisan. 2010. *Hay-Pay: Les verbs de l'arménien occidental*. Paris: L'Asiatheque.

Boyacioglu, Nisan & Hossep Dolatian. 2020. Armenian Verbs: Paradigms and verb lists of Western Armenian conjugation classes. Zenodo. doi:10.5281/ZENODO.4397423.

Bybee, Joan & Sandra Thompson. 1997. Three frequency effects in syntax. In *Annual meeting of the berkeley linguistics society*, vol. 23 1, 378–388.

Bybee, Joan L. 1985. *Morphology: A study of the relation between meaning and form*. Amsterdam: John Benjamins. doi:10.1075/tsl.9.

Calabrese, Andrea & Laura Grestenberger. 2024. Distributed morphology and historical linguistics. *Manuscript. url: https://ling.auf.net/lingbuzz/008003* .

Chan, Erwin. 2008. *Structures and distributions in morphological learning*. Philadelphia, PA: University of Pennsylvania dissertation.

Clahsen, Harald & Monika Rothweiler. 1993. Inflectional rules in children's grammars: Evidence from German participles. In *Yearbook of morphology 1992*, 1–34. Springer.

Clahsen, Harald, Monika Rothweiler, Andreas Woest & Gary Marcus. 1992. Regular and irregular inflection in the acquisition of German noun plurals. *Cognition* 45. 225–255.

Cobo-Lewis, Alan B, Barbara Zurer Pearson, Rebecca E Eilers & Vivian C Umbel. 2002. *Effects of bilingualism and bilingual education on oral and written english skills: A multifactor study of standardized test outcomes*. na.

Cournane, Ailís. 2017. In defense of the child innovator. In Eric Mathieu & Robert Truswell (eds.), *Micro change and macro change in diachronic syntax*, 10–24. Oxford: Oxford University Press. doi:10.1093/oso/9780198747840.003.0002.

Dolatian, Hossep. 2024. *Adjarian's Armenian dialectology (1911): Translation and commentary* (Languages of the Caucasus 4). Berlin: Language Science Press. doi:10.5281/zenodo.14008766.

Dolatian, Hossep & Peter Guekguezian. 2022a. Derivational timing of morphomes: Canonicity and rule ordering in the Armenian aorist stem. *Morphology* 32(3). 317–357. doi:10.1007/s11525-022-09397-x.

Dolatian, Hossep & Peter Guekguezian. 2022b. Relativized locality: Phases and tiers in long-distance allomorphy in Armenian. *Linguistic Inquiry* 1–41. doi:10.1162/ling_a_00456.

Dolatian, Hossep, Afsheen Sharifzadeh & Bert Vaux. 2023. *A grammar of Iranian Armenian: Parskahayeren or Iranahayeren* Languages of the Caucasus. Berlin: Language Science Press.

Donabedian, Anaid. 2016. The aorist in Modern Armenian: Core values and contextual meanings. In Zlatka Guentchéva (ed.), *Aspectuality and temporality: Descriptive and theoretical issues*, 375–412. Amsterdam: John Benjamins Publishing Company. doi:10.1075/slcs.172.12don.

Doyle, A_, Mireille Champagne & Norman Segalowitz. 1978. Some issues in the assessment of linguistic consequences of early bilingualism. *Aspects of bilingualism* 13–20.

Dresher, B Elan & Aditi Lahiri. 2022. The foot in the history of english: Challenges to metrical coherence. In Patrick Honeybone, Graeme Trousdale, Bettelou Los & Clare Cowie (eds.), *English historical linguistics: Change in structure and meaning*, 42–59. John Benjamins.

Dum-Tragut, Jasmine. 2009. *Armenian: Modern Eastern Armenian*, vol. 14 London Oriental and African Language Library. Amsterdam: John Benjamins Publishing Company. doi:10.1075/loall.14.

Embick, David. 2015. *The morpheme: A theoretical introduction*. Boston and Berlin: Walter de Gruyter. doi:10.1515/9781501502569.

Emond, Emeryse & Rushen Shi. 2020. Infants' rule generalization is governed by the Tolerance Principle. In *45th annual boston university conference on language development*, .

Fenson, Larry, Philip S Dale, J Steven Reznick, Elizabeth Bates, Donna J Thal, Stephen J Pethick, Michael Tomasello, Carolyn B Mervis & Joan Stiles. 1994. Variability in early communicative development. *Monographs of the Society for Research in Child Development* i–185.

van Gelderen, Elly. 2011. *The linguistic cycle: Language change and the language faculty*. Oxford University Press.

Godel, Robert. 1975. *An introduction to the study of Classical Armenian*. Wiesbaden: Reichert Wiesbaden.

Goodman, Judith C, Philip S Dale & Ping Li. 2008. Does frequency count? Parental input and the acquisition of vocabulary. *Journal of Child Language* 35(3). 515–531. doi:10.1017/s0305000907008641.

Greppin, John AC. 1973. The origin of Armenian nasal suffix verbs. *Zeitschrift für vergleichende Sprachforschung* 87(2. H). 190–198.

Guekguezian, Peter Ara & Hossep Dolatian. 2024. Distributing theme vowels across roots, verbalizers, and voice in Western Armenian verbs. In Robert Autry, Gabriela de la Cruz, Luis A. Irizarry Figueroa, Kristina Mihajlovic, Tianyi Ni, Ryan Smith & Heidi Harley (eds.), *Proceedings of the 39th West Coast Conference on Formal Linguistics*, 313–321. Somerville, MA: Cascadilla Proceedings Project.

Halle, Morris. 1962. Phonology in generative grammar. *Word* 18(1-3). 54–72. doi:10.1080/00437956.1962.11659765.

Halle, Morris & Alec Marantz. 1993. Distributed morphology and the pieces of inflection. In Kenneth Hale & Samuel J. Keyser (eds.), *The view from Building 20: Studies in linguistics in honor of Sylvaln Bromberger*, 111–176. Cambridge, MA: The MIT Press.

Hamp, Eric P. 1975. On the nasal presents of Armenian. *Zeitschrift für vergleichende Sprachforschung* 89(1. H). 100–109.

Hare, Mary & Jeffrey L Elman. 1995. Learning and morphological change. *Cognition* 56(1). 61–98.

Hart, Betty & Todd R Risley. 1995. *Meaningful differences in the everyday experience of young American children*. Baltimore, MD: Paul H Brookes Publishing. doi:10.5860/choice.33-3435.

Hart, Betty & Todd R Risley. 2003. The early catastrophe: The 30 million word gap by age 3. *American Educator* 27(1). 4–9.

Henke, Ryan E. 2023. Rules and exceptions: A tolerance principle account of the possessive suffix in northern east cree. *Journal of Child Language* 50(5). 1119–1154.

Hock, Hans Henrich. 2003. Analogical change. *The handbook of historical linguistics* 441–460.

Hockett, Charles F. 1942. A system of descriptive phonology. *Language* 18(1). 3–21. doi: 10.2307/409073.

Irani, Ava. 2019. *Learning from positive evidence: The acquisition of verb argument structure*: University of Pennsylvania dissertation.

Janse, Mark. 2009. Watkins' law and the development of agglutinative inflections in asia minor greek. *Journal of Greek Linguistics* 9(1).

Jelinek, Frederick. 1997. *Statistical methods for speech recognition*. MIT press.

Karakaş, Ayla, Hossep Dolatian & Peter Guekguezian. 2021. Effects of zero morphology on syncretism and allomorphy in Western Armenian verbs. In *Proceedings of the Workshop on Turkic and Languages in Contact with Turkic*, vol. 6, 5056. doi:10.3765/ptu.v6i1.5056.

Kenstowicz, Michael. 1996. Base-identity and uniform exponence: Alternatives to cyclicity. In Jacques Durand & Bernard Laks (eds.), *Current trends in phonology: Models and methods*, 363–393. Salford: University of Salford.

Kern, Sophie. 2003. Le compte-rendu parental au service de l'évaluation de la production lexicale des enfants français entre 16 et 30 mois. *Glossa* (85). 48–61.

Khanjian, Hrayr. 2013. *(Negative) concord and head directionality in Western Armenian*: Massachusetts Institute of Technology dissertation.

Khurshudian, Victoria G., Misha A. Daniel, Dmitri V. Levonian, Vladimir A. Plungian, Alex E. Polyakov & Sergey A. Rubakov. 2009. Eastern Armenian National Corpus. In *Computational Linguistics and Intellectual Technologies (Papers from the Annual International Conference "Dialogue 2009"*, vol. 8, 509–518. Moscow: RGGU.

Kim, Ronald. 2018. The prehistory of the Classical Armenian weak aorist. *Acta Linguistica Petropolitana* 14(1). 86–136. doi:10.30842/alp2306573714104.

Kiparsky, Paul. 1973. Abstractness, Opacity, and Global Rules. In Osamu Fujimura (ed.), *Three dimensions of linguistic theory*, Tokyo: Taikusha.

Kocharov, Petr. 2019. *Old Armenian nasal verbs. Archaisms and innovations*: Leiden University dissertation.

Kodner, Jordan. 2019. Estimating child linguistic experience from historical corpora. *Glossa: a journal of general linguistics* 4(1).

Kodner, Jordan. 2020. *Language Acquisition in the Past*: University of Pennsylvania dissertation.

Kodner, Jordan. 2023a. *Laissez-Faire* Analogical Change. *Journal of Historical Syntax* 7(6-19).

Kodner, Jordan. 2023b. What learning Latin verbal morphology tells us about morphological theory. *Natural Language & Linguistic Theory* .

Kodner, Jordan & Caitlin Richter. 2020. Transparent /ai/-raising as a contact phenomenon. In Ruaridh Purse & Yosiane White (eds.), *Penn working papers in linguistics: Selected papers from nwav*, vol. 25 2, 61–70.

Koulaguina, Elena & Rushen Shi. 2019. Rule generalization from inconsistent input in early infancy. *Language Acquisition* 26(4). 416–435.

Kroch, Anthony. 2005. Modeling language change and language acquisition. In *Expansion of an lsa institute forum lecture*, .

Labov, William. 1989. The child as linguistic historian. *Language Variation and Change* 1(1). 85–97. doi:10.1017/s0954394500000120.

Labov, William. 2007. Transmission and diffusion. *Language* 83(2). 344–387.

Labov, William. 2011. *Principles of linguistic change, volume 3: Cognitive and cultural factors*, vol. 36. John Wiley & Sons.

Labov, William, Malcah Yaeger & Richard Steiner. 1972. *A quantitative study of sound change in progress*, vol. 1. US Regional Survey.

Li, Daoxin & Kathryn D. Schuler. 2023. Acquiring recursive structures through distributional learning. *Language Acquisition* 30(3-4). 323–336. doi:10.1080/10489223.2023.2185522.

Lightfoot, David W. 1979. *Principles of diachronic syntax*. Cambridge: Cambridge University Press.

Lignos, Constantine & Charles Yang. 2018. Morphology and language acquisition. *Cambridge handbook of morphology* 765–791.

Loukatou, Georgia, Camila Scaff, Katherine Demuth, Alejandrina Cristia & Naomi Havron. 2021. Child-directed and overheard input from different speakers in two distinct cultures. *Journal of Child Language* .

MacWhinney, Brian. 1978. The acquisition of morphophonology. *Monographs of the Society for Research in Child Development* 1–123. doi:10.2307/1166047.

Maratsos, Michael. 2000. More overregularizations after all: New data and discussion on Marcus, Pinker, Ullman, Hollander, Rosen & Xu. *Journal of Child Language* 27(1). 183–212. doi:10.1017/s0305000999004067.

Marcus, Gary F, Steven Pinker, Michael Ullman, Michelle Hollander, T John Rosen, Fei Xu & Harald Clahsen. 1992. Overregularization in language acquisition. *Monographs of the society for research in child development* .

Martirosyan, Hrach. 2018. The Armenian dialects. In Geoffrey Haig & Geoffrey Khan (eds.), *The Languages and Linguistics of Western Asia*, 46–105. Berlin/Boston: De Gruyter. doi: 10.1515/9783110421682-003.

Martirosyan, Hrach. 2019. The Armenian dialects: Archaisms and innovations; description of individual dialects. *Bulletin of Armenian Studies / Армянский гуманитарный вестник* 5. 164–258.

Maxudianz, Mesrop. 1911. *Le parler arménien d'Akn*. Paris: Libraire Paul Geuthner. `https://books.google.com/books?id=IOffAAAAMAAJ`.

Mayol, Laia. 2007. Acquisition of irregular patterns in Spanish verbal morphology. In Ville Nurmi & Dmitry Sustretov (eds.), *Proceedings of the twelfth ESSLLI Student Session*, 1–11. Dublin.

Miller, George A. 1957. Some effects of intermittent silence. *The American journal of psychology* 70(2). 311–314.

Miller, Karen. 2013. Acquisition of variable rules:/s/-lenition in the speech of Chilean Spanish-speaking children and their caregivers. *Language Variation and Change* 25(3). 311–340.

Milroy, James & Lesley Milroy. 1985. Linguistic change, social network and speaker innovation1. *Journal of linguistics* 21(2). 339–384.

Nagy, William E & Richard C Anderson. 1984. How many words are there in printed school English? *Reading Research Quarterly* 304–330. doi:10.2307/747823.

Nardy, Aurélie, Jean-Pierre Chevrot & Stéphanie Barbu. 2014. Sociolinguistic convergence and social interactions within a group of preschoolers: A longitudinal study. *Language Variation and Change* 26(3). 273–301.

Nercissians, Emilia. 2001. Bilingualism and diglossia: Patterns of language use by ethnic minorities in Tehran. *International Journal of the Sociology of Language* 2001(148). 59–70. doi:10.1515/ijsl.2001.014.

Nercissians, Emilia. 2012. Life and culture of Armenians in Iran. *Language Discourse & Society* 2. 31–54.

Niyogi, Partha & Robert C Berwick. 1997. A dynamical systems model for language change. *Complex Systems* 11(3). 161–204.

Nowenstein, Iris, Sigríðður Sigurjónsdóttir, Charles Yang, Anton Ingason, Joel Wallenberg et al. 2020. The meaning of case: Morphosyntactic bootstrapping and icelandic datives. In *45th annual boston university conference on language development*, Cascadilla Press.

Olsen, Birgit Anette. 2017. The morphology of Armenian. In Jared Klein, Brian Joseph & Matthias Fritz (eds.), *Handbook of Comparative and Historical Indo-European Linguistics*, 1080–1097. Berlin/Munich/Boston: Walter de Gruyter. doi:10.1515/9783110523874-017.

Paul, Hermann. 1880. *Prinzipien der sprachgeschichte*. T ubingen. Niemeyer.

Payne, Sarah & Charles Yang. 2023. Making good on bads: Commentary on dressler et al. 2020. *Italian Journal of Linguistics* .

Pierrehumbert, Janet B. 2003. On frequency, transparency and productivity. In *Probabilistic linguistics*, 177–228. Cambridge, MA: MIT Press.

Richter, Caitlin. 2021. *Alternation-sensitive phoneme learning: Implications for children's development and language change*: University of Pennsylvania dissertation.

Ringe, Don & Charles Yang. 2022. The threshold of productivity and the 'irregularization' of verbs in Early Modern English. In *English historical linguistics: Change in structure and meaning*, 91. John Benjamins.

Roberts, Julie. 1997. Acquisition of variable rules: a study of (-t, d) deletion in preschool children. *Journal of child language* 24(2). 351–372.

Rydland, Veslemøy, Vibeke Grøver & Joshua Lawrence. 2014. The potentials and challenges of learning words from peers in preschool: A longitudinal study of second language learners in norway. In *Children's peer talk: Learning from each other*, 214–234. Cambridge University Press Cambridge.

Schuler, Kathryn D. 2017. *The acquisition of productive rules in child and adult language learners*: Georgetown University dissertation.

Shi, Rushen & Emeryse Emond. 2023. The threshold of rule productivity in infants. *Frontiers in Psychology* 14. 1251124. doi:10.3389/fpsyg.2023.1251124.

Smith, Jennifer, Mercedes Durham & Liane Fortune. 2009. Universal and dialect-specific pathways of acquisition: Caregivers, children, and t/d deletion. *Language Variation and Change* 21(1). 69–95.

Sneller, Betsy, Josef Fruehwald & Charles Yang. 2019. Using the Tolerance Principle to predict phonological change. *Language Variation and Change* 31(1). 1–20.

Stump, Gregory T. 2001. *Inflectional morphology: A theory of paradigm structure*. Cambridge: Cambridge University Press. doi:10.1017/CBO9780511486333.

Stump, Gregory T. 2006. Heteroclisis and Paradigm Linkage. *Language* 82(2). 279–322. doi:10.1353/lan.2006.0110.

Szagun, Gisela, Claudia Steinbrink, Melanie Franik & Barbara Stumper. 2006. Development of vocabulary and grammar in young German-speaking children assessed with a German language development inventory. *First Language* 26(3). 259–280. doi:10.1177/0142723706056475.

Thomson, Robert W. 1989. *An introduction to Classical Armenian*. Delmar, NY: Caravan Books.

Trips, Carola & Thomas Rainsford. 2022. Tolerating subject-experiencers? Yang's Tolerance Principle applied to psych verbs under contact in Middle English. *Journal of Historical Syntax* 6(12). 1–43.

Vaux, Bert. 1998. *The phonology of Armenian*. Oxford: Clarendon Press.

Vaux, Bert. 2012. The Armenian dialect of Smyrna. In Richard Hovannisian (ed.), *Armenian Smyrna/Izmir: The Aegean Communities*, 111–126. Costa Mesa, CA: Mazda Publishers.

Vaux, Bert. in prep. The Armenian dialect of New Julfa, Isfahan. Unpublished manuscript.

Watkins, Calvert. 1962. *Indo-european origins of the celtic verb. 1: The sigmatic aorist*. Dublin Institute for Advanced Studies.

Weinreich, Uriel, William Labov & Marvin Herzog. 1968. Empirical foundations for a theory of language change. In W. Lehmann & Y. Malkiel (eds.), *Directions for historical linguistics*, 95–195. University of Texas Press.

Xu, Fei & Steven Pinker. 1995. Weird past tense forms. *Journal of Child Language* 22(3). 531–556. doi:10.1017/s0305000900009946.

Yang, Charles. 2002. *Knowledge and learning in natural language*. Oxford: Oxford University Press.

Yang, Charles. 2005. On productivity. *Linguistic Variation Yearbook* 5. 265–302. doi:10.1075/livy.5.09yan.

Yang, Charles. 2013. Who's afraid of George Kingsley Zipf? or: Do children and chimps have language? *Significance* 10(6). 29–34.

Yang, Charles. 2016. *The price of linguistic productivity*. Cambridge, MA: MIT Press. doi:10.7551/mitpress/9780262035323.001.0001.

Yang, Charles. 2021. A user's guide to the Tolerance Principle. *Manuscript. url: https://ling.auf.net/lingbuzz/004146* .

Zeman, Daniel, Joakim Nivre & Mitchell Abrams... 2021. Universal dependencies 2.9. LINDAT/CLARIAH-CZ digital library at the Institute of Formal and Applied Linguistics (ÚFAL), Faculty of Mathematics and Physics, Charles University. `http://hdl.handle.net/11234/1-4611`.

Томсон, Александр. 1887. *Лингвистические Исследования [Linguistic research]*, vol. 1. С.-Петербургъ: Тип. Е. Евдокимова.

Ալավերդյան, Ս. Ա. 1968. Արդվինի Բարբառի Ձևաբանության Մի Քանի Առանձնահատկությունները [Some features of the morphology of the dialect of Artvin]. *Պատմա-բանասիրական հանդես* 3. 229–240.

Աղայան, Էդուարդ Բագրատի. 1954. *Մեղրու Բարբառը [The dialect of Meghri]*. Երևան: Հայկական ՍՍՀ Գիտությունների Ակադեմիա Հրատարակչություն.

Աճառեան, Հրաչեայ. 1898. *Ուսումնասիրթիւններ Հայկական Գաւառաբանութեան*. Վենետիկ: Ս. Ղազար.

Աճառեան, Հրաչեայ. 1901. *Քննություն Ղարաբաղի Բարբառի [Study of the dialect of Karabakh]*. Վաղարշապատ: Տպարան Մայր Աթոռաց Ս Էջմիածնի.

Աճառեան, Հրաչեայ. 1911. *Հայ Բարբառագիտութիւն [Armenian dialectology]*. Մոսկուա-Նոր-Նախիջեւան}: Լազարեան Ճեմարան Արեւելեան Լեզուաց.

Աճառեան, Հրաչեայ. 1935. *Քննութիւն Ագուլիսի Բարբառի [Study of the dialect of Agulis]*. Երեւան: Պետական Հրատարակչութիւն.

Աճառեան, Հրաչեայ. 2021. *Քննութիւն Նոր-Նախիջեւանի (Խրիմի) Բարբառի [Examination of New Nakhichevan (Crimea) dialect*. Երեւան: ՀՀ ԳԱԱ Հ. Աճառյանի Անվան Լեզվի Ինստիտուտ.

Աճառյան, Հրաչյա. 1926. *Քննություն Մարաղայի Բարբառի [Examination of the Maragha dialect]*.

Աճառյան, Հրաչյա. 1940. *Քննություն Նոր-Ջուղայի Բարբառի [Examination of the New Julfa dialect]*.

Աճառյան, Հրաչյա. 1941. *Քննություն Պոլսահայ Բարբառի [Study of the Armenian dialect of Istanbul]*. Երևան: Պետական Համալսարանի Հրատարակչություն.

Աճառյան, Հրաչյա. 1947. *Քննություն Համշենի Բարբառի [Study of the dialect of Hamshen]*. Երևան: ՀՍՍՀ Գիտությունների Ակադեմիայի Հրատարակչություն.

Աճառյան, Հրաչյա. 1952. *Քննություն Վանի Բարբառը [Study of the the dialect of Van]*. Երևան: Երևանի Պետական Համալսարանի Հրատարակչություն.

Աճառյան, Հրաչյա. 1959. *Լիակատար Քերականություն Հայոց Լեզվի [Complete grammar of the Armenian language]*, vol. 4.1. Երևան: Հայկական ՍՍՀ Գիտությունների Ակադեմիա Հրատարակչություն.

Աճառյան, Հրաչյա. 1961. *Լիակատար Քերականություն Հայոց Լեզվի [Complete grammar of the Armenian language]*, vol. 4.2. Երևան: Հայկական ՍՍՀ Գիտությունների Ակադեմիա Հրատարակչություն.

Աճառյան, Հրաչյա. 2003. *Քննություն Կիլիկիայի Բարբառի [Study of the dialect of Cilicia]*. Երևան: Երևանի Համալսարանի Հրատարակչություն.

Անթոսյան, Սամուել Մինասի. 1961. *Կեսարիայի Բարբառը [The dialect of Kayseri]*. Երևան: Հայկական ՍՍՀ ԳԱ Հրատարակչություն.

Ասատրյան, Մանվել. 1962. *Ուրմիայի (Խոյի) Բարբառը [Dialect of Urmia or Khoy]*. Երևան: Երևանի Պետական Համալսարանի Հրատարակչություն.

Ասատրյան, Մանվել Եզորի. 1968. *Լոռու Խոսվածքը [The speech of Lori]*. Երևան: «Միտք» Հրատարակչություն.

Ավետյան, Սարգիս. 2020. Համաբանական Փոփոխությունների Երկու Միտում Արդի Արևելահայերենի Աորիստի Հարացույցում (Համաժամանակյա և Տարաժամանակյա Քննություն) [Two tendencies of analogical changes in the aorist paradigm of Modern Eastern Armenian (A synchronic and diachronic examination)]. *Բանբեր Երևանի Համալսարանի. Բանասիրություն* 31(1). 24–39.

Բաղրամյան, Ռ. Հ. 1976. Խոտրջուրի Բարբառը [The dialect of Khodorchur]. *Պատմա-բանասիրական հանդես* 3. 95–105.

Բաղրամյան, Ռ.Հ. 1960. *Դերսիմի Բարբառային Քարտեզը*. Երեւան: Հայկական ՍՍՀ Գիտությունների Ակադեմիա Հրատարակչություն.

Բաղրամյան, Ռուբեն. 1984. Նուզգերի Խոսվածքը [The vernacular of Nuzger]. *Լրաբեր հասարակական գիտությունների* 11(11). 40–47.

Գրիգորյան, Ա. 1957. *Հայ Բարբառագիտության Դասընթաց [Armenian dialectology course]*. Երեւան: Երեւանի Պետական Համալսարանի Հրատարակչություն.

Գևորգյան, Գայանե. 2016. Նիկոմեդիայի Խոսվածքի Ընդհանուր Բնութագիրը [General characteristics of the speech of Nicomedia]. In *Հայերենի Բարբառագիտական Ատլաս: Պրակ 4*, 39–59. «Աստղիկ» հրատարակչություն.

Գևորգյան, Գայանե. 2017. Ադափազարի Խոսվածքի Բարբառային Յուրահատկությունների Ընդհանուր Բնութագիրը [General characteristics of the dialectal peculiarities of Adapazar speech]. In Վ. Լ. Կատվալյան, Գ. Կ. Խաչատրյան & Վ. Գ. Համբարձումյան (eds.), *Գիտական Հոդվածների Ժողովածու. Ժողովածուն Նվիրվում Է Գրիգոր Ղափանցյանի Ծննդյան 130-Ամյակին*, 88–109. Երևան: «Աստղիկ».

Դանիելյան, Թևան. 1967. *Մալաթիայի Բարբառը [The dialect of Malatya]*. Երևան: Հայկական ՍՍՀ Գիտությունների Ակադեմիա Հրատարակչություն.

Դավթյան, Կարո Ստեփանի. 1966. *Լեռնային Ղարաբաղի Բարբառային Քարտեզը [Dialectal map of Mountainous Karabakh]*. Երևան: Հայկական ՍՍՀ Գիտությունների Ակադեմիայի Հրատարակչություն.

Թումայեան, Կարապետ. 1930. *Մարզուանի Գաւառաբարը [Provincial dialect of Merzifon]*. Վիեննա: Մխիթարեան Տպարան.

Խաչատրյան, Հ. Ս. 1985. Շապին Գարահիսարի Բարբառ (Հոլովումը և Խոնարհումը) [Dialect of Şebinkarahisar (declension and conjugation)]. In *Հայերենի Բարբառագիտական Ատլաս*, vol. 2, Երեւան: Հայկական ՍՍՀ ԳԱ Հրատարակչություն.

Խաչատրյան, Հասմիկ. 2016. Եվդոկիայի Միջբարբառը [The interdialect of Evdokia]. In *Հայերենի Բարբառագիտական Ատլաս: Պրակ 4*, 60–94. «Աստղիկ».

Կատվալյան, Վիկտոր. 2016. *Բայազետի Բարբառը և Նրա Լեզվական Առնչությունները Շրջակա Բարբառների Հետ [The dialect of Bayazet and its linguistic relationships with surrounding dialects]*. Երևան: «Աստղիկ» Հրատարակչություն.

Կատվալյանը, Վ. Լ., Ա. Ա. Աբրահամյանը, Ձ. Ա. Բառնասյանը & Հ. Ռ. Խաչատրյանը. 2016. *Գրական Ար Եմտահայերենի և ՀՀ Տարածքում Գործառող Կը Ճյուղի Բարբառների Առնչությունները [The relationships between the literary Western Armenian language and the Կը branch dialects in the territory of the Republic of Armenia]*. Երեւան: ՀՀ ԳԱԱ Հրաչյա Աճառյանի անվան լեզվի ինստիտուտ.

Կոստանդյան, Դարիկո Մեսրոպի. 1979. *Երզնկայի Բարբառը [The dialect of Yerznka]*. Երեւան: Հայկական ՍՍՀ ԳԱ Հրատարակչություն.

Հանեյան, Անահիտ Նշանի. 1978. *Տիգրանակերտի Բարբառը [The dialect of Tigranakert]*. Երևան: Հայկական ՍՍՀ ԳԱ Հրատարակչություն.

Ղազարյան, Բաբկեն Կիրակոսի. 1982. *Ժամանակակից Հայոց Լեզվի Հաճախականության Բառարան: Մոտավորապես 36200 Բառ [Frequency dictionary of modern Armenian: Approximately 36200 words]*. Երևան: Հայկական ՍՍՀ Գիտությունների Ակադեմիա Հրատարակչություն.

Ղարագյուլյան, Թերեզա. 1981. Ակնարկներ Ժամանակակից Հայերեն Խոսակցական Լեզվի [Observations on modern colloquial Armenian]. In Էդուարդ Բագրատի Աղայան (ed.), *Ժամանակակից Հայերենի Խոսակցական Լեզուն [Modern colloquial Armenian]*, 120–278. Երևան: Հայկական ՍՍՀ Գիտությունների Ակադեմիա.

Ղարիբյան, Ա. Ս. 1958. *Հայերենի Նորահայտ Բարբառների Մի Նոր Խումբ [A new group of newly discovered dialects of Armenian]*. Երեւան: Հայկական ՍՍՀ Գիտությունների Ակադեմիա Հրատարակչություն.

Ղարիբյան, Արարատ. 1941. *Համառոտություն Հայ Բարբառագիտության [Summary of Armenian dialectology]*. Երևան: Պետհամալսարանի տպ.. `https://arar.sci.am/dlibra/publication/307074/`.

Մադաթյան, Քերոբ Մկրտչի. 1985. *Ալաշկերտի Խոսվածքը [Alashkert speech]*. Երեւան: Հայկական ՍՍՀ ԳԱ Հրատարակչություն.

Մարգարյան, Ալեքսանդր Սիմոնի. 1975. *Գորիսի Բարբառը [The dialect of Goris]*. Երևան: Երևանի Համալսարնի Հրատարակչություն.

Մարկոսյան, Ռազմիկ Արարատի. 1989. *Արարատյան Բարբառ [The dialect of Ararat]*. Երևան: Լույս.

Մեժունց, Բարմիկ Խաչատուրի. 1989. *Շամշադին-Դիլիջանի Խոսվածքը [Shamshadin-Dilijan speech]*. Երեւան: Երեւանի Համալսարանի Հրատարակչութիւն.

Մեսրոպյան, Հայկանուշ. 2016. Ռոդոսթոյի Կամ Բուլղարահայերի Բարբառի Ընդհանուր Նկարագիրը [A general description of the dialect of Rodosto or Bulgarian Armenians]. In *Հայերենի Բարբառագիտական Ատլաս: Պրակ 4*, 95–118. «Աստղիկ» հրատարակչություն.

Մկրտչյան, Հակոբ Մկրտչի. 1952. *Կարնո Բարբառը (Հնչյունաբանություն, Ձևաբանություն, Բառարան) [The dialect Karin (phonology, morphology, and dictionary)]*. Երևան: Հայկական ՍՍՀ Գիտությունների Ակադեմիա Հրատարակչություն.

Մկրտչյան, Ներսես. 1971. *Բուրդուրի Բարբառը [The dialect of Burdur]*. Երևան: Հայկական ՍՍՀ ԳԱ Հրատարակչություն.

Մկրտչյան, Ներսես Ալեքսանի. 2006. *Անատոլիայի Նորահայտ Հայ Բարբառները և Բանահյուսությունը [Newly discovered Armenian dialects and folklore of Anatolia]*. Երեւան: Ա. հ.

Մուրադյան, Մարլենա Հայրիկի. 1962. *Շատախի Բարբառը [The dialect of Shatakh]*. Երևան: Երևանի Պետական Համալսարանի Հրատարակչություն.

Չոլաքեան, Յակոբ. 2009. *Քեսապի Բարբառը [The dialect of Kesab]*. Երեւան: ԵՊՀ Հրատարակչութիւն.

Պետոյան, Վարդան Ավետիսի. 1954. *Սասունի Բարբառը [The dialect of Sasun]*. Երևան: Հայկական ՍՍՀ Գիտությունների Ակադեմիա.

Ջահուկյան, Գևորգ Բեգլարի. 1972. *Հայ Բարբառագիտության Ներածություն [Introduction to Armenian dialectology]*. Երևան: ՀՀ ԳԱԱ Հրաչյա Աճառյանի անվան լեզվի ինստիտուտ.

Դաւիթ-Բէկ, Մելիք Ս. 1919. *Արաբկիրի Գաւառաբարբառը [The provincial dialect of Arapgir]*. Վիէննա: Մխիթարեանց տպարան.